Physically Constrained Fusion of Sentinel-1 and ERA5-Land for Daily 500-m Snow-Depth Retrieval Across Northern Hemisphere Mountains

Cheng Zhang, Lingmei Jiang, *Member, IEEE*, and Jiancheng Shi

***Abstract*—Quantifying the spatiotemporal distribution of mountain snowpack is critical for hydrology, as it represents a vital freshwater resource for over a billion people. While C-band synthetic aperture radar (SAR) from Sentinel-1 provides high-resolution estimates over mountains, its use for daily snow monitoring is hindered by infrequent revisits and signal attenuation in wet snow. To address these limitations, this work proposes the adaptive snow-phenology shape model fitting (ASP-SMF), a physically constrained fusion algorithm that generates all-weather continuous daily 500-m snow-depth maps for Northern Hemisphere mountains. Designed for data-scarce regions, ASP-SMF integrates Sentinel-1, ERA5-Land, and Interactive Multisensor Snow and Ice Mapping System (IMS) data independently of in situ calibration. The algorithm first adaptively segments the highly dynamic snow time series into three distinct phenological phases using physical rules of snowfall and snowmelt. Subsequently, it utilizes Sentinel-1 retrievals and IMS snow-free constraints to correct the temporal, amplitude, and offset biases of the reanalysis data via a robust reduced major axis (RMA) regression and uncertainty-aware sequential least-squares programming (SLSQP) optimization. Comprehensive validation against 565 sites and 55 airborne LiDAR/photogrammetry surveys (2017–2021) demonstrates that ASP-SMF: 1) reduces errors relative to Sentinel-1 under dry snow conditions, reducing root-mean-square error (RMSE) by 26% (from 0.27 to 0.20 m) and increasing R^2 from 0.49 to 0.72; 2) exhibits pronounced improvement under wet snow conditions, reducing RMSE by 23% (from 0.61 to 0.47 m) and boosting R^2 by 145% (from 0.22 to 0.54); and 3) shows comparable performance to the SNOw Data Assimilation System (SNODAS) product without in situ data assimilation, with a higher Kling–Gupta efficiency (KGE) (0.57 versus 0.37) against Airborne Snow Observatory (ASO) LiDAR in the Rocky Mountains. Critically, this independence from in situ inputs enables ASP-SMF to deliver spatiotemporally consistent snow monitoring in ungauged regions, providing a useful framework for high-resolution assessment of mountain snow water resources.**

***Index Terms*—Data fusion, mountain hydrology, Sentinel-1, snow depth, snow phenology, synthetic aperture radar (SAR).**

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I. Introduction

MOUNTAIN snowpacks, often termed natural water towers, are a fundamental component of the Northern Hemisphere's hydrological system and climate regulation. They provide essential freshwater for over a billion people [1], supply approximately three-fourths of crop irrigation water in the western United States [2], and are critical for renewable hydroelectric energy production [3].

Despite its profound importance, accurately quantifying snow depth across vast and topographically complex mountain ranges remains a significant challenge. Ground-based station networks offer valuable, high-frequency point measurements, yet they are too sparsely distributed to adequately represent the pronounced spatial heterogeneity of mountain snowpacks [4]. Although remote sensing technologies provide the necessary spatial coverage, currently operational sensors face distinct limitations. Passive microwave sensors offer global snow depth, but are constrained by their coarse spatial resolution (~25 km) and signal saturation in snowpacks deeper than approximately 0.8 m [6], [7]. Considerable research has therefore focused on downscaling or modeling snow depth using snow redistribution schemes, optical remote sensing, passive microwave observations, and machine learning (ML) approaches [8], [9], [10], [11], [12]. In parallel, numerical weather prediction, land-surface, snowpack, and hydrologic modeling frameworks have also been applied to snow estimation and prediction, including snow height, snow water equivalent, and snow-cover evolution in mountainous regions [13], [14]. Recent studies have further explored hybrid strategies that combine physical model outputs, ML algorithms, and climate forcing information for snow and hydrologic prediction. [15], [16], [17]. Together, these data-driven and model-based approaches provide valuable alternatives for snow-depth estimation, snowpack prediction, and multisource fusion. However, their application in remote mountain regions can still be affected by the availability and representativeness of training samples, predictor datasets, meteorological forcing, model resolution, and region-specific calibration. Conversely, active microwave observations from synthetic aperture radar (SAR) have emerged as a promising solution for mapping snow depth at high spatial resolution over large scales. The change detection method, based on the sensitivity of the C-band (5.4 GHz) cross-polarization ratio (VH/VV) from the Sentinel-1 (S1) constellation to snow depth, has proven

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effective for retrieving mountain snow depth [6], [18], [19]. Given its high spatial resolution, these retrievals are increasingly utilized as training data for ML models [5], [12] or assimilated into land surface models [20], [21]. However, the applicability of C-band SAR for continuous monitoring is constrained by three principal limitations.

First, SAR-based snow-depth retrievals often exhibit significant uncertainties. This arises because the complex relationship between the backscatter signal and snow depth depends on the snow properties [22], which makes robust physical inversion difficult and purely empirical approaches prone to errors [23], [24], [25]. Second, while Sentinel-1 excels at identifying wet snow [26], the presence of liquid water strongly attenuates the C-band signal. This attenuation decouples the relationship between backscatter and snow depth, leading to severe underestimation biases during the spring ablation period [27]. Third, the effective temporal resolution of satellite SAR is insufficient to capture rapid snow dynamics. The necessity of masking wet snow severely compromises the theoretical 6- to 12-day revisit period, often resulting in sparse data coverage—for instance, dropping to fewer than two observations per month during the critical melt season of April in the Sierra Nevada (see Appendix B).

Given this temporal sparsity of SAR, reanalysis products offer a viable alternative for providing the daily temporal information required for continuous monitoring. These datasets leverage land surface models—such as the hydrology tiled ECMWF scheme for surface exchanges over land (HTESSEL) model within ERA5-Land—to simulate physical snow processes including accumulation, melt, and refreezing, driven by meteorological forcing derived from data assimilation [28]. Among available reanalysis products, ERA5-Land has been shown to provide more accurate mountain snow-depth estimates than its counterparts like ERA5, JRA-55, and MERRA-2 [29], [30], [31]. However, it is constrained by two primary limitations. First, its 9-km spatial resolution, while an improvement over the 31 km of the parent ERA5 product, is still too coarse to resolve the complex topography and fine-scale spatial patterns of mountain landscapes. Second, studies have documented notable inaccuracies in ERA5-Land snow depth [32], [33], particularly a delayed onset of the ablation process during the melt season [29].

Existing multisource snow products and correction methods partly address these challenges, but they remain limited for generating daily sub-kilometer snow-depth fields over data-scarce mountain regions. Passive-microwave-based hemispheric products, such as GlobSnow SWE, provide valuable long-term snow water equivalent estimates, but GlobSnow SWE is limited by its coarse spatial resolution and the exclusion of mountainous regions [34], [35]. Regional operational systems such as SNOw Data Assimilation System (SNODAS) provide daily high-resolution snow analyses by assimilating multiple observational sources into a snow modeling framework, but their transferability is constrained by the need for data-intensive regional assimilation infrastructure and dense observational support [36], [37], [38]. Recent fusion methods that combine passive microwave and optical data with ML techniques have improved the spatial detail of snow-depth estimates by incorporating snow-cover, terrain, meteorological, or external training information [5], [9], [11], [29], [39]. However, these methods often rely on cloud-free optical composites, in situ data, or region-specific calibration, which limits their direct applicability to data-scarce mountain regions. Similarly, reanalysis bias-correction and downscaling studies can improve ERA5 or ERA5-Land snow estimates in complex terrain [40], but they generally do not directly address how to reconstruct daily high-resolution snow-depth trajectories when SAR retrievals are temporally sparse and degraded during wet-snow conditions.

To bridge this gap, this study aims to develop a robust method for generating large-scale daily 500-m snow-depth maps in data-scarce mountain regions. The proposed framework is designed to combine the high spatial resolution of Sentinel-1 snow-depth retrievals with the daily temporal continuity of ERA5-Land, while reducing the influence of Sentinel-1 temporal gaps, wet-snow retrieval limitations, and systematic timing and magnitude biases in the reanalysis time series. Our approach builds upon the concept of shape model fitting (SMF), which provides accurate temporal evolution information without relying on ancillary in situ data or extensive training sample libraries [41]. While originally developed for vegetation remote sensing [42], [43], [44], this methodology has demonstrated higher accuracy and flexibility than traditional change-based fusion algorithms such as the spatial and temporal adaptive reflectance fusion model (STARFM) [45].

However, directly applying the SMF framework to mountain snow depth presents two principal challenges. First, unlike the predictable, cyclical patterns of vegetation phenology [43], [46], [47], snowpack evolution is highly stochastic, driven by episodic storms and rapid melt. This invalidates the use of standard temporal templates and necessitates a more flexible framework capable of adapting to the distinct physical phases of snow accumulation and ablation. Second, both the Sentinel-1 and ERA5-Land input datasets contain significant, well-documented biases in timing and magnitude. To address these dual uncertainties, our approach introduces two key innovations: 1) it incorporates adaptive temporal shifts to correct timing mismatches and 2) it employs a reduced major axis (RMA) regression to account for errors in both variables, thereby avoiding over-reliance on either potentially flawed data source. Compared with fully predictive ML frameworks, this design prioritizes interpretability and transferability by using physically interpretable correction terms and open-access observational anchors, without requiring in situ calibration.

Consequently, this study proposes the adaptive snow-phenology SMF (ASP-SMF), a novel physically constrained fusion algorithm designed to generate daily, seamless 500-m snow-depth maps for Northern Hemisphere mountains. Unlike traditional methods, this approach adaptively segments the highly dynamic snow time series into distinct phenological phases using physical rules of snowfall and snowmelt. By implementing an uncertainty-aware optimization strategy, it effectively overcomes the limitations of C-band SAR signal attenuation and operates using exclusively open-access data, including Sentinel-1, ERA5-Land, and Interactive Multisensor

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TABLE I
INPUT DATASETS FOR THE ASP-SMF ALGORITHM

Dataset	Resolution	Variable(s) Used	Data Source
ERA5-Land	9 km	Snow depth, snowfall, snowmelt	Copernicus CDS
Sentinel-1 Snow Depth	500 m	Snow depth, wet snow flag	C-SNOW Portal
IMS	1 km	Snow cover (snow-free class)	NSIDC DAAC

Snow and Ice Mapping System (IMS) data, without requiring any in situ observations. The remainder of this article is organized as follows. Section II describes the study area and datasets used for implementation and validation. Section III details the proposed ASP-SMF algorithm. Section IV presents a comprehensive evaluation of the product against airborne LiDAR, photogrammetry, and in situ station data across the Northern Hemisphere. Finally, Section V discusses the technical nuances and uncertainties, followed by a conclusion in Section VI.

II. DATA AND STUDY AREAS

A. Model Input Data and Preprocessing

The ASP-SMF algorithm relies solely on three open-access datasets as input, summarized in Table I. The core of the algorithm synergizes the temporally continuous daily dynamics from the ERA5-Land reanalysis with the spatially precise anchor points provided by the high-resolution Sentinel-1 (S1) snow-depth product. In addition, the IMS data are used to determine reliable snow-free areas.

We selected the daily, aggregated 9-km ERA5-Land reanalysis [48] due to its documented superior performance in mountainous regions [30], [31]. The 500-m snow-depth product [6], [18] served as the input source, with the wet snow flag mask applied to strictly select observations from dry snow conditions. The daily 1-km IMS product [49] was employed, providing zero-snow-depth information to impose necessary constraints on the algorithm in snow-free areas.

All datasets were spatiotemporally aligned to the daily 500-m Sentinel-1 grid before fusion. ERA5-Land snow depth was resampled using cubic spline interpolation, which provides a smooth interpolation of continuous geophysical fields and has been used in previous snow, cryosphere, and reanalysis-remote sensing data-merging studies [50], [51], [52]. In contrast, the categorical IMS snow-cover product was resampled using the nearest-neighbor method to preserve its original snow/no-snow classes. This preprocessing step ensured consistent grid geometry, projection, spatial extent, and daily temporal matching among the input datasets.

B. Validation Data and Representative Regions

To rigorously assess the generalizability and accuracy of the ASP-SMF product, we compiled a comprehensive suite of independent reference datasets across four distinct Northern Hemisphere mountain ranges (see Fig. 1). These regions were selected to cover a diverse spectrum of snow regimes, spanning the maritime Sierra Nevada, the continental Rocky Mountains, the transitional European Alps, and the data-scarce Karakoram. These datasets, summarized in Table II, facilitate a multifaceted evaluation, including: 1) spatial validation against high-resolution airborne remote sensing products; 2) point-scale evaluation against in situ station networks; and 3) intercomparison with a benchmark operational product.

For high-resolution spatial validation in North America, we used 50 airborne LiDAR flights from the Airborne Snow Observatory (ASO), which are considered a benchmark with a reported mean absolute error (MAE) of less than 8 cm against manual measurements [2]. In European Alps, the primary reference was high-precision snow-depth maps derived from five airborne photogrammetry surveys over Davos region, a widely used validation standard with a reported root-mean-square error (RMSE) of approximately 0.15–0.25 m [53], [54].

Point-scale evaluation was performed using a total of 565 independent stations from three networks: the Global Historical Climatology Network (GHCN), incorporating the SNOTEL network in USA [55], [56]; the high-elevation Intercantonal Measurement and Information System (IMIS) in the Alps [57]; and stations from the China Meteorological Administration (CMA) in the Karakoram. Critically, our algorithm was not calibrated against any of these in situ observations, ensuring a truly independent assessment.

Finally, we evaluated the new algorithm against the SNODAS, a leading operational system in the contiguous United States [36], [37]. This intercomparison is particularly meaningful as SNODAS represents a data-intensive, regionally tuned approach, whereas ASP-SMF is an in situ-independent, globally applicable algorithm. All airborne and SNODAS datasets were resampled to the 500-m target grid using cubic spline interpolation prior to analysis.

III. METHODS

The ASP-SMF is a physically constrained fusion algorithm developed to generate daily, seamless 500-m snow-depth maps for mountainous regions. Here, "physically constrained" means that the fusion procedure is guided by snow-process-informed constraints, including snow-phenology-based segmentation, physically interpretable correction terms for amplitude, timing, and offset biases, and snow-free anchoring from IMS. These constraints are used to maintain physically plausible snow-depth trajectories while integrating the temporal continuity of ERA5-Land reanalysis with the high-resolution spatial detail of Sentinel-1 (S1) snow-depth observations.

The algorithm consists of two main stages. First, we perform an adaptive segmentation of the snow season into distinct phases to account for the distinct physical characteristics governing the relationship between reanalysis and satellite observations. Second, a phase-specific SMF is applied to correct the amplitude, temporal, and offset biases in the ERA5-Land time series, using S1 and IMS snow-cover data as robust anchors. This two-stage approach is specifically designed to address the rapid and stochastic nature of mountain

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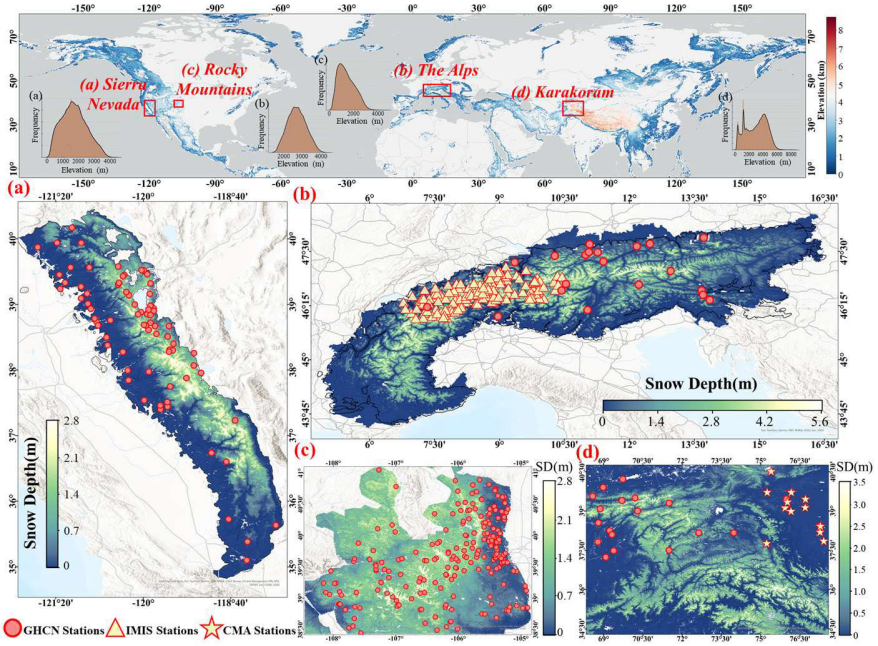


Fig. 1. Overview of the four primary study areas. (Top) Global locations of (a) Sierra Nevada, (b) Alps, (c) Rocky Mountains, and (d) Karakoram on a map of high-mountain terrain [73], with corresponding elevation distributions. (Bottom) Detailed views of each region, overlaid with mean winter (December–February, 2017–2021) Sentinel-1 snow depth [18] and validation station locations (GHCN, IMIS, and CMA).

TABLE II
VALIDATION DATASETS

Data Type	Dataset Name	Study Area(s)	Key Specifications	Extent
Airborne	Airborne Snow Observatory (ASO) LiDAR	Sierra Nevada, Rockies	3 m resolution; MAE < 8 cm	50 flights (2017–2021)
	Airborne Photogrammetry	European Alps (Davos)	0.5 m resolution; RMSE ~0.2 m	5 surveys (2017–2021)
In-situ	GHCN (incl. SNOTEL)	N. America, Alps, Karakoram	Point measurements	411 stations
	IMIS Network	European Alps	Point measurements	142 stations
	CMA Stations	Karakoram	Point measurements	12 stations
Benchmark Product	SNODAS	Sierra Nevada, Rockies	1 km resolution; daily maps	Daily (2017–2021)

snowpack evolution. The overall workflow is illustrated in Fig. 2.

A. Stage 1: Physically Adaptive Snow Phenology Segmentation

The biases and relationship between ERA5-Land and S1 observations are not static but state-dependent. Consequently, a single, season-long correction model would fail to capture the unique error characteristics associated with different physical processes (e.g., accumulation versus melt). Therefore, instead of applying a uniform correction, we first segment the snow time series into distinct phenological phases. Unlike the predictable, gradual cycles of vegetation phenology, mountain snowpack evolution is highly stochastic, driven by episodic storms and rapid melt events. Our segmentation scheme is designed to capture this dynamic behavior through a two-step process: an initial fine-scale segmentation followed by a temporal regularization.

1) Initial Classification via Adaptive Thresholding: The initial classification assigns each pixel-day to one of four states based on the 3-day change rate of the ERA5-Land snow-depth time series. We use ERA5-Land for this step because, while its absolute snow-depth values may be biased, its physically based land surface model provides a spatiotemporally continuous and dynamically realistic representation of snow accumulation and ablation trends. The four initial states are delineated using regionally adaptive thresholds (detailed in Appendix A) as follows.

1) *No-Snow Phase:* It is characterized by negligible snow depth and minimal change rates. This phase represents periods before seasonal accumulation or after complete

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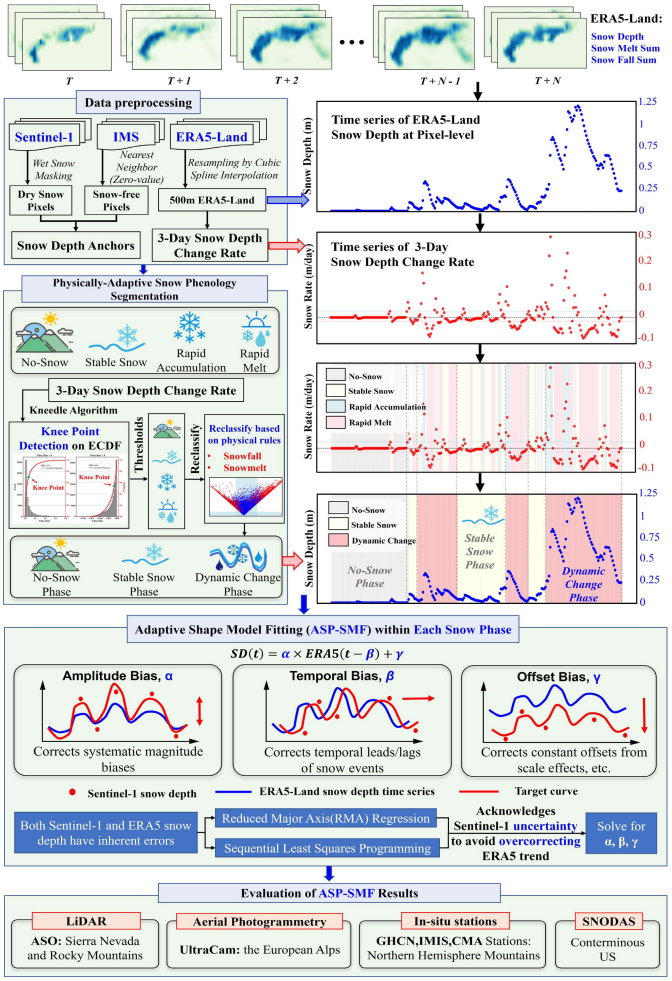


Fig. 2. Schematic workflow of the ASP-SMF algorithm. The process integrates multisource data preprocessing, adaptive snow phenology classification using the Kneedle algorithm, and phase-specific SMF to reconstruct daily, spatially continuous 500-m snow depth.

melt, where the surface energy balance is not dominated by snow.

2) *Stable Snow Phase:* It represents periods with a consolidated snowpack subject to slow processes such as compaction, sublimation, and minor precipitation or melt events. These processes result in low-magnitude changes in snow depth.
3) *Rapid Accumulation Phase:* It is characterized by high-magnitude positive changes in snow depth, driven by significant synoptic-scale weather events, such as major snowfall from cyclonic storms.
4) *Rapid Melt Phase:* It is characterized by high-magnitude negative changes in snow depth, typically driven by strong warming events, high solar radiation, or rain-on-snow events.

To objectively distinguish between stable and rapid phases, we avoid fixed, subjective thresholds, which lack generality across different climate regimes. Instead, we employ a data-driven, regionally adaptive approach using the Kneedle algorithm [58]. For each study region, we compute the empirical cumulative distribution function (ECDF) of all positive (accumulation) and negative (melt) 3-day change rates from the ERA5-Land data. The "knee point" on the ECDF curve, detected by the Kneedle algorithm, marks the optimal statistical transition from high-frequency, low-magnitude background changes (the stable snow phase) to low-frequency, high-magnitude events (the rapid phases). This process yields two adaptive thresholds for each mountain range—one for accumulation and one for melt. This ensures our segmentation is statistically optimized for the local snow dynamics and enhances the algorithm's objectivity and reproducibility.

2) Phase Consolidation and Temporal Regularization: The initial daily segmentation results in a fragmented time series with many short-lived segments (e.g., 1–3 days), which is unsuitable for robust SMF. To address this, we developed a two-part consolidation and regularization process to generate temporally coherent and physically meaningful phases.

First, we consolidate the four initial states into three functionally distinct phases: no-snow phase, stable snow phase, and dynamic change phase. Here, the rapid accumulation and rapid melt phases are merged into the single dynamic change phase. This merger is physically and algorithmically motivated. Physically, both precursor phases represent periods of intense mass and energy exchange between the snowpack and the atmosphere, marking the most unstable periods of the snow season. Algorithmically, they share the common challenge of accurately fitting a steep temporal gradient, whether positive or negative.

Second, to filter out transient noise and ensure sufficient segment length for analysis, we enforce a minimum duration of six days. This threshold was chosen to balance two requirements. Algorithmically, a six-day window provides sufficient data points to define a nonlinear shape, enabling robust fitting and preventing the misinterpretation of noise as a trend. Physically, this duration aligns with the typical 3-to-7-day timescale over which synoptic-scale weather systems, such as mid-latitude cyclones, evolve through their lifecycle [59], ensuring that each segment corresponds to a persistent meteorological event rather than daily weather variability.

To implement this consolidation, we developed an iterative temporal consolidation algorithm. This algorithm systematically reclassifies short-lived segments by evaluating their temporal context and underlying physical drivers. It employs a set of hierarchical rules based on majority assimilation within a dynamic window and a conditional re-evaluation using ERA5-Land fluxes. A detailed description of this algorithm and its logic is provided in Appendix A.

The final output of this stage is a continuous, pixelwise time series that classifies each day into one of the three core phases (no-snow, stable snow, or dynamic change), setting a robust foundation for the adaptive SMF described next.

B. Stage 2: ASP-SMF Within Each Phase

Following phase classification, the ASP-SMF algorithm performs pixelwise SMF within each continuous phase segment. The core principle is to correct the systematic biases in coarse-resolution, temporally continuous ERA5-Land snow depth (SD_{ERA5}) by fitting its temporal shape to sparse, high-resolution anchor observations (SD_{anchor}). This anchor dataset

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combines the quantitative retrievals from Sentinel-1 (SD_{S1}) with explicit zero-snow-depth constraints (1 = 0) from IMS. The strategy for integrating IMS and the validation of its high-precision snow-free detection are detailed in Appendix C.

Recognizing that error characteristics are state-dependent, a unique set of correction parameters is derived for each individual phase segment, ensuring the correction is tailored to the specific snowpack conditions.

1) Spatiotemporal Correction Model: We hypothesize that within any given phase, the relationship between the predicted true snow depth and the ERA5-Land model output can be approximated by a linear transformation that accounts for three primary sources of systematic bias: 1) amplitude bias (α), where ERA5-Land systematically overestimate or underestimate the magnitude of accumulation or melt events, corrected by a multiplicative scaling factor α; 2) temporal bias (β), where ERA5-Land's modeled snow events may lead or lag the actual ground truth, corrected by a time-shift parameter β (in days); and 3) offset bias (γ), where a constant numerical offset exists between datasets due to factors like scale effects, corrected by an additive term γ.

This leads to the core spatiotemporal correction model

$$SD_{pred}(t) = \alpha \cdot SD_{ERA5}(t-\beta) + \gamma \tag{1}$$

where $SD_{pred}(t)$ is the final corrected daily snow depth. The objective of the ASP-SMF algorithm is to find the optimal parameter set (α, β, and γ) that best fits the corrected ERA5-Land time series to the available SD_{anchor} points within each phase segment. In this formulation, the three correction terms have explicit physical interpretations: α adjusts the magnitude of snow-depth variation, β accounts for possible timing mismatches in accumulation or ablation, and γ represents an additive offset. The optimization is constrained to keep these corrections within physically plausible ranges, thereby preserving realistic snow-depth evolution while correcting systematic biases in ERA5-Land.

2) Dual-Strategy Parameter Optimization: Robust fitting requires a sufficient number of anchor points; therefore, the ASP-SMF algorithm employs a dual-strategy approach adaptable to the data availability within each phase segment. The selection of the optimization strategy is determined by the count of valid anchor observations (M).

In phase segments where sufficient anchor points are available ($M \geq 3$), correction parameters are derived via a grid search that minimizes the RMSE between predicted snow depths and anchor observations. The threshold of three observations was selected to ensure that valid data cover at least half of the minimum six-day phase duration defined in Section III-A, thereby providing sufficient statistical weight to constrain the model. The search operates within a physically constrained parameter space to ensure realism: the amplitude scaling factor is bounded to $\alpha \in [0.3, 2.0]$ to prevent extreme artifacts, and the temporal shift is limited to $\beta \in [-6, 6]$ days, consistent with the typical duration of synoptic-scale weather systems [59].

For each parameter combination, the optimal parameter set $(\alpha_{opt}, \beta_{opt}, \gamma_{opt})$ that minimizes the RMSE between the predicted snow depth (SD_{pred}) and the anchor observations (SD_{anchor}) is thus identified as

$$(\alpha_{opt}, \beta_{opt}, \gamma_{opt}) = argmin_{\alpha,\beta,\gamma} \sqrt{\frac{1}{M}\sum_{i=1}^{M}(SD_{anchor} - SD_{pred,i})^2}. \tag{2}$$

Conversely, in data-sparse conditions where a specific phase contains limited anchors ($M < 3$), which is common during prolonged wet snow periods, minimizing RMSE directly risks overfitting to sparse and potentially uncertain Sentinel-1 retrievals. To mitigate this, we implemented a robust nested optimization framework that utilizes anchor points from the entire snow season. This approach iterates through potential temporal shifts ($\beta \in [-6, 6]$) while solving for amplitude and offset using RMA regression coupled with sequential least-squares programming (SLSQP). This approach explicitly addresses the dual uncertainty inherent in both the model simulation (ERA5-Land) and satellite retrievals (Sentinel-1). Unlike ordinary least squares (OLS), which assumes the independent variable is error-free, RMA minimizes the orthogonal distance to the fit line, thereby symmetrically treating uncertainties in both datasets. This prevents the algorithm from over-relying on limited anchor points and better preserves the trend information from ERA5-Land. Furthermore, SLSQP enforces the same physical constraints described above to preclude mathematically optimal but physically implausible solutions (e.g., negative amplitude factors).

For each discrete temporal shift β, the optimal α and γ are determined by solving the following constrained minimization problem using available anchors (N) in the snow season:

$$(\alpha_{opt}, \gamma_{opt}) = argmin_{\alpha,\gamma} \sum_{i=1}^{N} \frac{(\alpha X_i - Y_i + \gamma)^2}{\alpha^2 + 1} \tag{3}$$

subject to $0.3 \leq \alpha \leq 2.0$ and $-2.0 \leq \gamma \leq 2.0$, where X_i and Y_i represent the time-aligned pairs of SD_{ERA5} and SD_{anchor}, respectively. Finally, the algorithm evaluates the candidate parameter triplets generated from all potential temporal shifts and selects the set yielding the lowest RMSE against the sparse anchors as the final optimal solution $(\alpha_{opt}, \beta_{opt}, \gamma_{opt})$. In extremely rare instances where valid anchor data are insufficient to support the RMA-SLSQP convergence, the algorithm defaults to using the spatially resampled 500-m ERA5-Land snow depth to fill the gap. The complete Python implementation of the ASP-SMF algorithm, including the optimization modules, is open-source and available (see Data Availability).

C. Performance Evaluation

The performance of the ASP-SMF product was comprehensively evaluated against airborne remote sensing data (LiDAR and photogrammetry), extensive in situ station networks, and the operational SNODAS product. For robust comparison, all spatial datasets were resampled to the 500-m target grid using cubic spline interpolation. For the in situ data, in cases where multiple stations fell within a single 500-m grid cell, their

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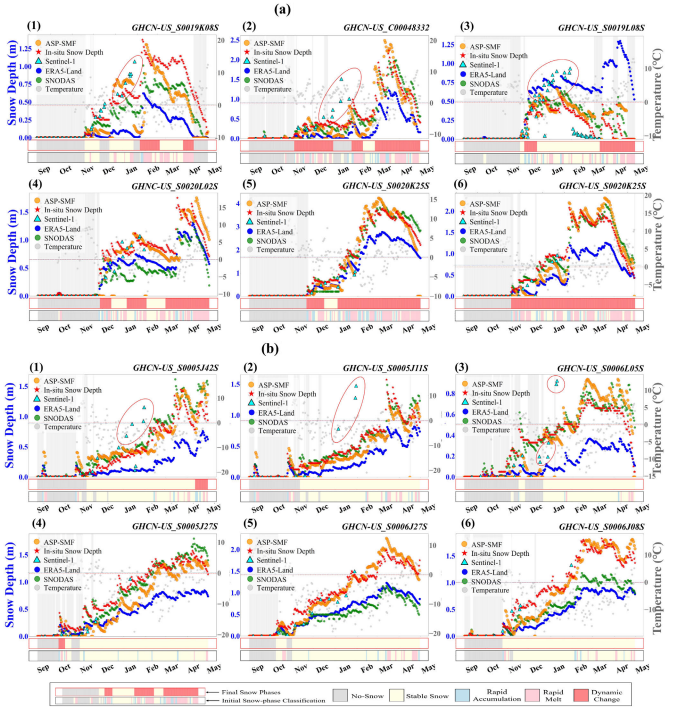


Fig. 3. Daily snow-depth time series at selected sites in (a) Sierra Nevada and (b) Rocky Mountains for the 2020 water year. ASP-SMF estimates are compared with in situ data, input datasets (ERA5-Land and S1), SNODAS, and air temperature. The bars at the bottom show (bottom bar) initial and (top bar) final consolidated snow phase classifications. Red ellipses highlight periods of particular interest discussed in the text.

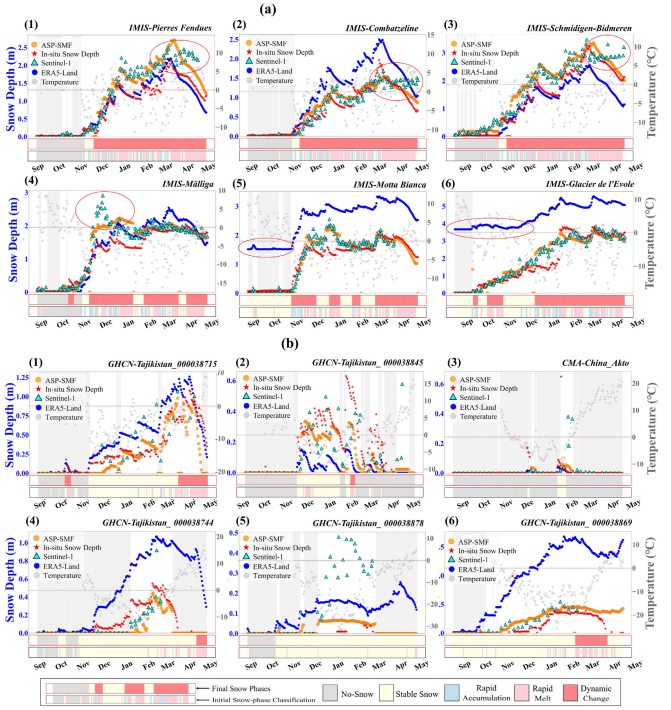


Fig. 4. Daily snow-depth time series at selected sites in (a) European Alps and (b) Karakoram for the 2020 water year. ASP-SMF estimates are compared with in situ data, input datasets (ERA5-Land and Sentinel-1), and air temperature. The bars at the bottom show (bottom bar) initial and (top bar) final consolidated snow phase classifications. Red ellipses highlight periods of particular interest discussed in the text.

daily snow-depth values were averaged to create a single, representative observation for that grid cell.

Performance was quantified using a suite of metrics. The product's ability to classify snow cover was assessed using accuracy, precision, and recall. The accuracy of continuous snow-depth values was evaluated with the RMSE, MAE, bias, coefficient of determination R^2, and Spearman's rank correlation coefficient (ρ). We also employed the Kling–Gupta efficiency (KGE) [60], which provides a composite assessment of correlation, bias, and variability, with a value of 1 indicating a perfect match.

IV. Results

A. Temporal Performance at In Situ Stations

To qualitatively assess the performance of the ASP-SMF algorithm, we first examine the daily snow-depth time series at representative in situ stations. This analysis highlights the algorithm's ability to address challenging scenarios, such as significant biases in the input data and prolonged periods of data scarcity. We first present the results from North America (see Fig. 3) before discussing the performance in Eurasia (see Fig. 4).

Fig. 3 demonstrates the performance of ASP-SMF at selected sites in the Sierra Nevada and Rocky Mountains, where the key challenges include sparse S1 retrievals during melt and comparison against the high-performing SNODAS product. In the Sierra Nevada, the algorithm consistently addresses these issues. For instance, at sites where S1 retrievals show a trend inconsistent with in situ data [see Fig. 3(a-2) and (a-3)], ASP-SMF is not unduly influenced and successfully tracks the ground truth, because the RMA regression treats S1 observations as uncertain anchors rather than absolute truth, allowing the underlying ERA5-Land trend to maintain stability. Furthermore, during the spring melt when wet snow conditions lead to a lack of S1 anchors [see Fig. 3(a-4)–(a-6)], the algorithm maintains a physically realistic ablation curve.

A key finding in these regions is the strong performance of the in situ-independent ASP-SMF relative to the data-assimilating SNODAS product. Across many sites in the Rocky Mountains [see Fig. 3(b)], ASP-SMF matches or exceeds the accuracy of SNODAS, particularly in capturing peak snow depth and the seasonal evolution more accurately where SNODAS sometimes exhibits an early melt bias [see Fig. 3(b-1) and (b-3)].

Fig. 4 showcases the algorithm's robustness in the distinct environments of European Alps, which feature frequent S1 observations but potentially biased reanalysis, and the Karakoram, a data-scarce, high-altitude region. In European Alps, where ERA5-Land often exhibits a significant high bias [see Fig. 4(a-5) and (a-6)], ASP-SMF successfully corrects this overestimation by anchoring the time series to the more reliable S1 observations, utilizing only the temporal pattern of the reanalysis.

The Karakoram sites test the algorithm's limits in an extremely data-sparse environment. Despite the infrequent S1 observations, ASP-SMF effectively constrains the time series.

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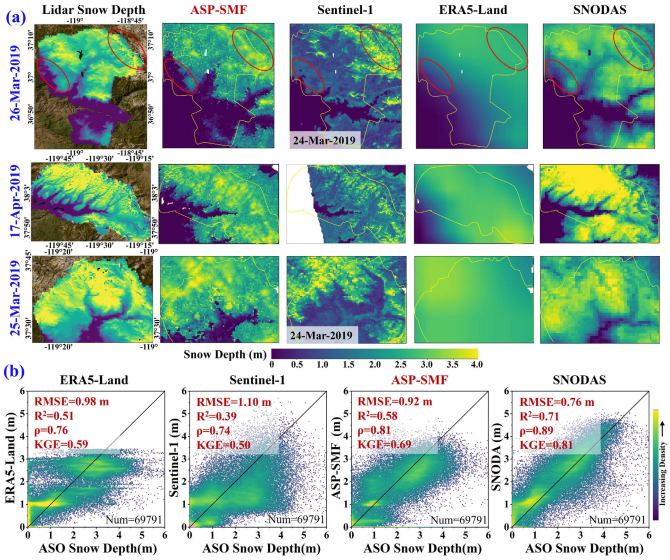


Fig. 5. Spatial validation against ASO LiDAR in the Sierra Nevada. (a) Maps comparing the ASO reference snow depth with ASP-SMF, Sentinel-1, ERA5-Land, and SNODAS for three different dates and locations. Red ellipses highlight key features. Sentinel-1 dates are noted when not coincident with the ASO flight. (b) Scatter density plots comparing each product against the ASO reference, aggregated over all coincident observations from 2017 to 2021.

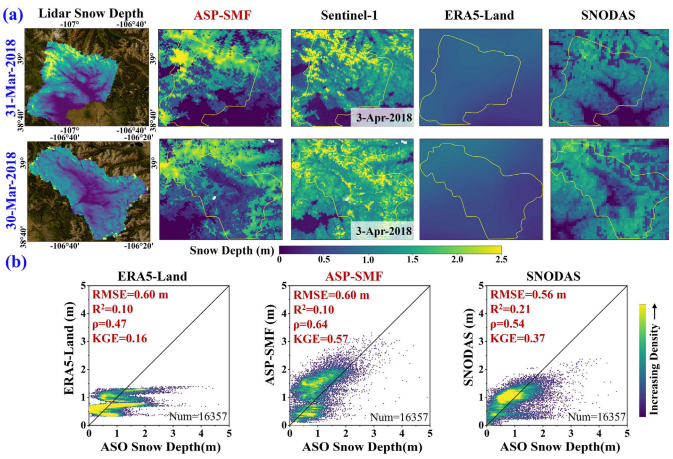


Fig. 6. Spatial validation against ASO LiDAR in the Rocky Mountains. (a) Map comparison for two dates. Sentinel-1 dates are noted when not coincident with the ASO flight. (b) Aggregated scatter density plots comparing ERA5-Land, Sentinel-1, ASP-SMF, and SNODAS against the ASO reference.

It successfully corrects a significant overestimation from a flawed S1 retrieval at one site [see Fig. 4(b-5)] and accurately captures the shallow snowpack dynamics where ERA5-Land failed at another [see Fig. 4(b-6)]. These results demonstrate the crucial role of the phase-based fitting logic in producing reliable estimates even with minimal anchor data.

B. Spatial Validation Against Airborne Reference Data

The spatial accuracy and pattern fidelity of the ASP-SMF product were rigorously evaluated against airborne reference datasets in both North America and Europe. This included high-resolution (3 m) LiDAR data from the ASO in U.S. mountains and high-precision (0.5 m) photogrammetry-derived snow-depth maps in European Alps.

1) Validation Against ASO LiDAR in U.S. Mountains: In the Sierra Nevada, ASP-SMF demonstrates a marked improvement over its input datasets and strong performance against the operational SNODAS product (see Fig. 5). Qualitatively, the spatial patterns of ASP-SMF show a strong correspondence with the ASO reference data, capturing fine-scale features such as deep snowpacks on high-elevation ridges and the transition to snow-free valley floors (see Fig. 5(a), first row). The algorithm successfully fills gaps in the Sentinel-1 data and corrects its underestimation bias during the melt season (see Fig. 5(a), second row). Quantitatively, the basin-wide scatter plot analysis, aggregating all available ASO flights from 2017 to 2021, confirms that ASP-SMF (RMSE = 0.92 m and R^2 = 0.58) substantially outperforms its Sentinel-1 anchor input (RMSE = 1.10 m and R^2 = 0.39).

These findings highlight a key achievement of the algorithm: it successfully synergizes sparse S1 anchors with a coarse reanalysis to produce a final product that is both spatially complete and quantitatively more accurate than its primary high-resolution input. At the same time, the validation against ASO LiDAR provides an absolute-error context for this improvement, reflecting the remaining uncertainty in deep and heterogeneous mountain snowpacks.

In the Rocky Mountains, ASP-SMF shows competitive advantages over established products, including SNODAS (see Fig. 6). Visually, ASP-SMF effectively captures high-elevation snow patterns where SNODAS shows a low bias and corrects for the tendency of Sentinel-1 to overestimate shallow snow in valley areas [see Fig. 6(a)]. The quantitative assessment reveals that ASP-SMF achieves the highest KGE (=0.57) and Spearman's rank correlation (ρ = 0.64), outperforming SNODAS (KGE = 0.37 and ρ = 0.54). While SNODAS exhibits a marginally lower RMSE (0.56 versus 0.60 m), ASP-SMF's higher KGE indicates a better overall representation of the snow-depth distribution and variability.

This result is particularly noteworthy, as it demonstrates that the globally applicable, in situ-independent ASP-SMF algorithm can achieve comparable overall performance relative to a regionally tuned, data-assimilating operational product like SNODAS in this continental snow environment.

2) Validation Against Airborne Photogrammetry in European Alps: In European Alps, the ASP-SMF product was validated against high-precision (0.5 m) snow-depth maps derived from airborne photogrammetry over Davos region (see Fig. 7). The spatial comparison reveals that ASP-SMF successfully reconstructs fine-scale snow distribution patterns, providing substantial detail beyond the coarse ERA5-Land and filling observational gaps in the Sentinel-1 product [see Fig. 7(a)]. A basin-averaged time-series analysis against in situ stations within the photogrammetry footprint shows that while ASP-SMF effectively tracks the seasonal evolution in 2018 and 2020, both the Sentinel-1 input and the final ASP-SMF product exhibited a significant low bias during the challenging 2019 water year. The potential reasons for this underperformance are analyzed in Section V-D.

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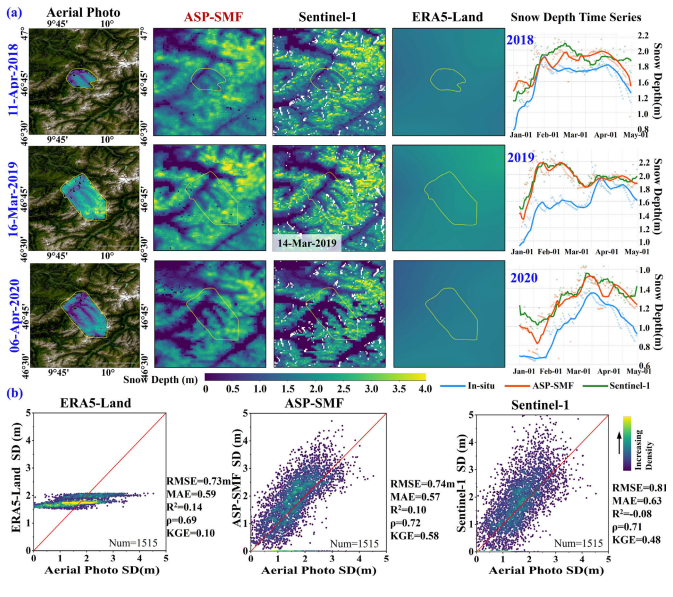


Fig. 7. Spatial validation against airborne photogrammetry in European Alps (Davos). (a) Map comparison for three dates (left four columns). The far-right column shows the basin-averaged daily time series of ASP-SMF, Sentinel-1, and in situ data for the corresponding years. (b) Aggregated scatter density plots comparing each product against the photogrammetry reference.

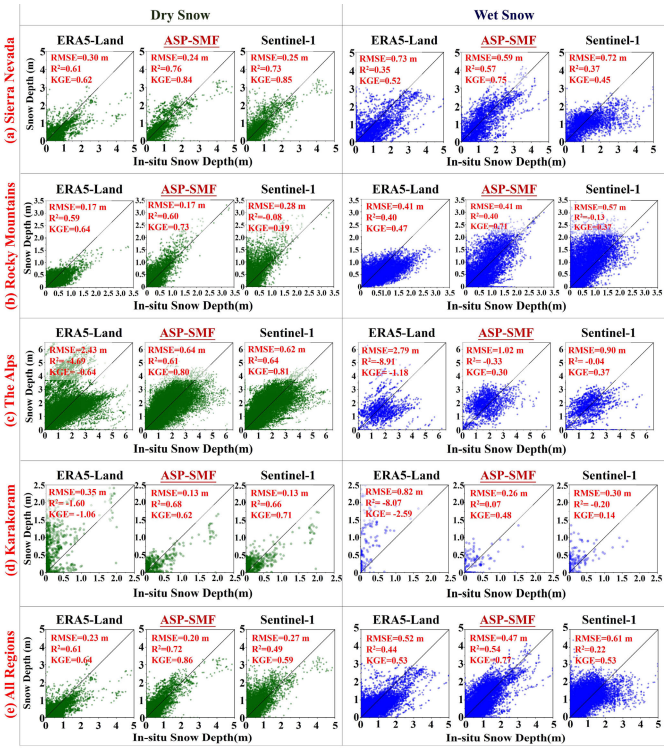


Fig. 8. Scatter density plots comparing daily snow depth from ERA5-Land, ASP-SMF, and Sentinel-1 against in situ station observations. The comparison is shown for (a)–(d) each of the four study regions individually and (e) all regions combined. Data are stratified into dry snow (green) and wet snow (blue) conditions. The analysis is performed on a common set of co-located observations where valid Sentinel-1 data exists.

The quantitative scatter plot analysis [see Fig. 7(b)] further confirms the algorithm's effectiveness. ASP-SMF (KGE = 0.58 and ρ = 0.72) demonstrates a significant improvement in accuracy over both its primary inputs, ERA5-Land (KGE = 0.10) and Sentinel-1 (KGE = 0.48 and ρ = 0.71). These robust metrics underscore the algorithm's ability to synergize two imperfect data sources to produce a final product that is more accurate than either of its individual components.

C. Point-Scale Evaluation Under Dry and Wet Snow Conditions

We conducted a comprehensive point-scale evaluation against in situ observations from the GHCN, IMIS, and CMA networks. To ensure a direct and fair comparison, the analysis was restricted to days where valid Sentinel-1 retrievals were available, ensuring all products were evaluated against an identical set of observation points. We used the Sentinel-1 wet snow flag to stratify the analysis into dry and wet snow conditions. Fig. 8 presents the results for each of the four mountain ranges individually (a)–(d) and for all regions combined (e).

The aggregated results from all stations [see Fig. 8(e)] provide robust confirmation of the ASP-SMF algorithm's improvement over its Sentinel-1 input. Under dry snow conditions, ASP-SMF achieves a significantly lower overall RMSE of 0.20 m and a higher KGE of 0.86, outperforming Sentinel-1 (RMSE = 0.27 m and KGE = 0.59). This performance advantage is even more pronounced under wet snow conditions. While Sentinel-1's accuracy degrades substantially (RMSE = 0.61 m and KGE = 0.53), ASP-SMF maintains robust performance (RMSE = 0.47 m and KGE = 0.77). This represents a 23% reduction in RMSE and a 45% improvement in KGE during these critical melt periods, effectively mitigating the known limitations of SAR-based snow-depth retrievals in wet snow.

This performance is competitive with representative regional snow-depth retrieval studies. Sentinel-1-based retrievals have reported strong dry-snow performance in European Alps [6], [18], whereas broader C-SNOW/Sentinel-1 evaluations over Northern Hemisphere mountains reported ubRMSE values of approximately 0.61–0.68 m and correlation coefficients of 0.52–0.58 [25]. Recent ML and fusion studies also provide useful benchmarks, including MAE = 0.33 m and R = 0.78 for Sentinel-1-based ML retrievals in European Alps [12], MAE = 0.17–0.33 m and R = 0.61–0.81 for optical-/passive-microwave mountain snow-depth estimation [5], and centimeter-scale RMSEs for optical-/passive-microwave fusion or downscaling products in shallower snow regimes over China and the Tibetan Plateau [8], [9], [11], [61], [62], [63]. Because these benchmarks differ in snow-depth dynamic range, regional snow climate, spatial resolution, validation reference, and training data dependence, they should be interpreted as contextual rather than strictly one-to-one comparisons. Within this context, ASP-SMF achieves competitive dry-snow accuracy (RMSE = 0.20 m and R^2 = 0.72) while providing a daily, seamless, 500-m product across diverse mountain regions without in situ calibration.

It should be noted, however, that the wet-snow RMSE of 0.47 m remains nonnegligible in absolute terms. This error level may still be considerable for applications requiring highly

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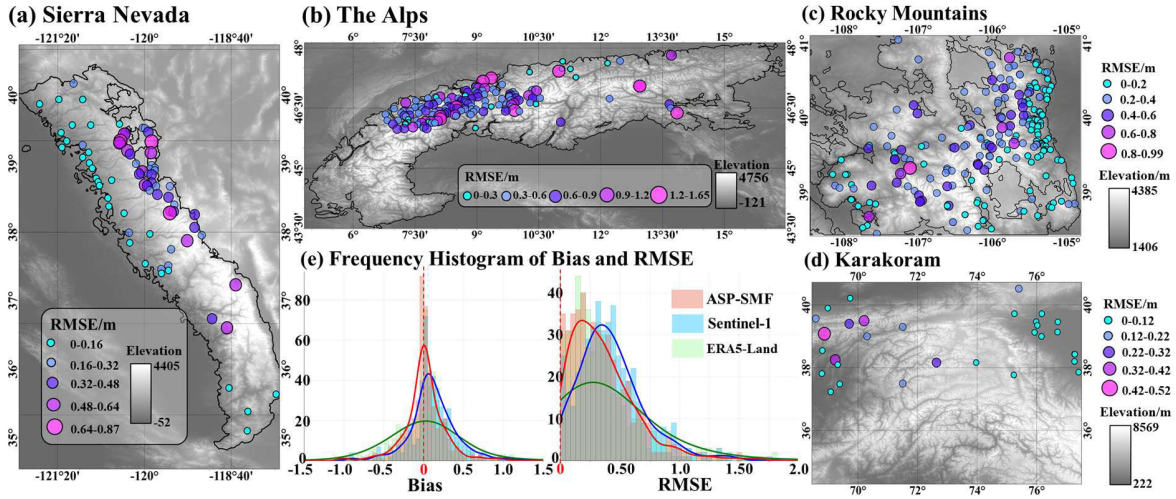


Fig. 9. Spatial distribution and frequency of errors against in situ stations (2017–2021). (a)–(d) Maps showing the time-averaged RMSE of the ASP-SMF product at each station location. (e) Frequency histograms of bias and RMSE for ASP-SMF, Sentinel-1, and ERA5-Land, aggregated across all stations in the four study regions.

accurate local snow-depth estimates or direct conversion of snow depth into water volume. Nevertheless, the wet-snow result should be interpreted in the context of one of the most difficult regimes for microwave snow-depth retrieval [6], [12], [23]. Under wet-snow conditions, ASP-SMF shows a small overall bias (Bias = −0.04 m) and a relatively strong rank consistency with in situ observations (Spearman's ρ = 0.74), indicating that the product preserves useful information on the timing and relative magnitude of snow-depth changes during the ablation period. Thus, the main value of ASP-SMF under wet snow lies in providing a continuous and physically plausible representation of melt-season evolution, rather than eliminating all absolute snow-depth uncertainty.

The individual regional results [see Fig. 8(a)–(d)] show that this overall pattern of improvement is generally consistent across different mountain climates, although regional differences remain. The enhanced robustness of ASP-SMF during wet snow is particularly evident in the North American ranges, whereas the Alps remain more challenging. Overall, the station-based evaluation provides quantitative evidence that ASP-SMF consistently matches or exceeds the accuracy of its Sentinel-1 anchor data under dry conditions and provides a more stable estimate during wet-snow periods.

To quantify metric uncertainty and assess whether the improvements were statistically significant across sites, we further performed a site-level uncertainty and significance analysis for the station-based validation. The detailed procedure, including station-level bootstrap confidence intervals (CIs) and paired site-level Wilcoxon signed-rank tests [64], is provided in Appendix F.

D. Spatial Distribution and Sources of Error

To further understand the performance of ASP-SMF, we analyzed the spatial distribution of errors and investigated the propagation of uncertainties from the input datasets into the final product.

1) Spatial Patterns of In Situ Validation Errors: The spatial distribution of the time-averaged RMSE at each in situ station reveals a distinct topographic relationship [see Fig. 9(a)–(d)]. Across all four mountain ranges, errors are generally lower in low-elevation regions and tend to increase at higher elevations. This pattern reflects the larger dynamic range of snow depth typically found at high altitudes. This elevation dependence is hydrologically relevant because high-elevation accumulation zones often dominate seasonal snow storage and runoff generation, highlighting the need to consider spatially variable uncertainty when using ASP-SMF for mountain water-resource assessment or snow-hydrological modeling. The aggregated frequency histograms of bias and RMSE from all stations provide a quantitative summary of the overall performance [see Fig. 9(e)]. The error distributions for ASP-SMF are visibly more concentrated around zero and exhibit smaller tails compared with both Sentinel-1 and ERA5-Land, confirming that ASP-SMF provides a more accurate and less biased estimate than either of its input datasets.

2) Error Propagation From Input Datasets: To apportion the error sources, we analyzed the correlation between the residuals (product minus in situ observation) of ASP-SMF and its input datasets (see Fig. 10). Positive correlations are observed between the residuals of ASP-SMF and Sentinel-1 across all regions, with the correlation coefficient (R) reaching as high as 0.90 in the Alps [see Fig. 10(c)]. This high correlation in the Alps (R = 0.90) suggests that in environments with dense, high-quality anchor data, the ASP-SMF algorithm effectively conforms to the S1 observations, faithfully reproducing their spatial patterns—including both their strengths and intrinsic uncertainties. In contrast, the correlation with ERA5-Land residuals is substantially weaker. This indicates that the accuracy of the Sentinel-1 anchor data is a primary driver of the final product's accuracy.

Crucially, the analysis also highlights the algorithm's robustness against large, anomalous errors in the reanalysis data. In the Alps and Karakoram, where ERA5-Land exhibits

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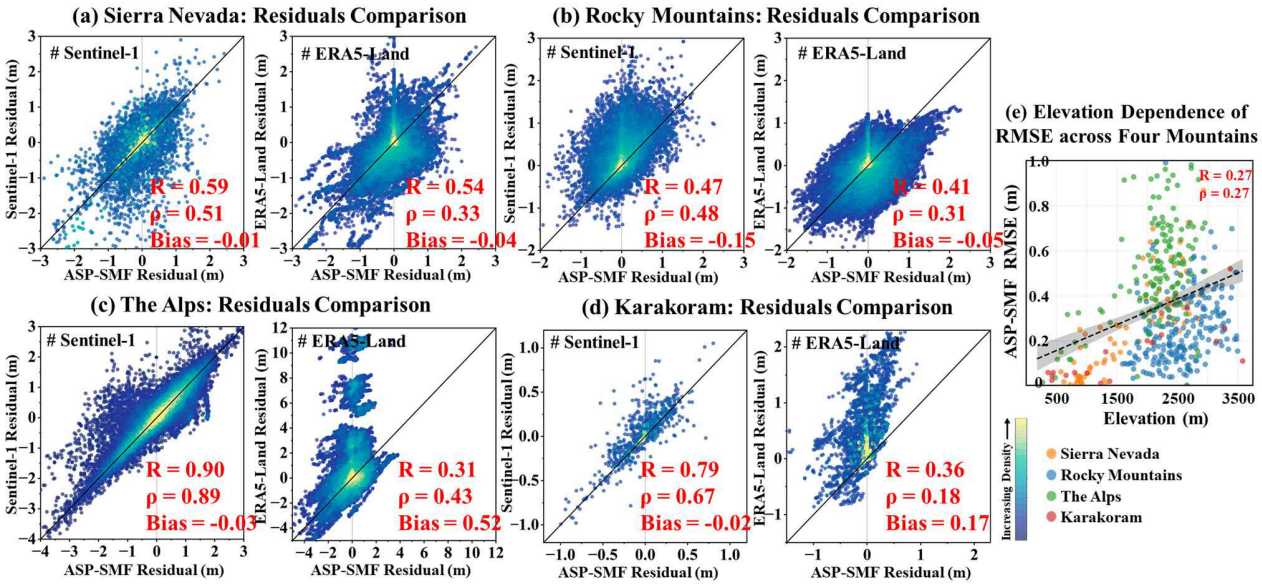


Fig. 10. Analysis of error propagation from input datasets. (a)–(d) Scatter density plots comparing the residuals of ASP-SMF (versus in situ) with the residuals of (left) Sentinel-1 and (right) ERA5-Land for each mountain range. (e) Relationship between ASP-SMF RMSE and station elevation across all sites.

significant, systematic biases [see Fig. 10(c) and (d)], these large errors do not propagate into the ASP-SMF product, whose residuals remain well-constrained. Finally, the analysis reveals a weak but statistically significant positive correlation (R = 0.27) between the RMSE of ASP-SMF and station elevation [see Fig. 10(e)].

In summary, the error analysis indicates that while ASP-SMF significantly reduces the overall error compared with its inputs, its accuracy is most strongly influenced by the quality of the Sentinel-1 anchor data. The algorithm effectively buffers against large biases in the ERA5-Land reanalysis, suggesting that further improvements to the final product are most likely to be achieved by enhancing the upstream Sentinel-1 snow-depth retrieval algorithm itself.

E. Spatiotemporal Performance and Long-Term Stability

1) Spatiotemporal Continuity and Dynamic Evolution: A key advantage of the ASP-SMF algorithm is the generation of a daily, spatially seamless 500-m snow-depth product that overcomes the substantial data gaps in the original Sentinel-1 retrievals. Fig. 11 visually demonstrates this capability, presenting 10-day interval snapshots of the snowpack evolution for the 2020–2021 water year in the Sierra Nevada and European Alps. Corresponding results for the Rocky Mountains and the Karakoram are provided in Appendix D.

The map sequences reveal several key findings. First, ASP-SMF produces a continuous and physically realistic depiction of the snowpack lifecycle, from winter accumulation to topographically driven ablation in spring. Second, on dates with valid Sentinel-1 data, ASP-SMF spatial patterns show high consistency with S1 retrievals, confirming that the algorithm effectively preserves the underlying high-resolution information. Third, and most critically, ASP-SMF successfully captures dynamic events missed by Sentinel-1's infrequent revisits. In the Alps (see Fig. 11(b), red boxes), for example,

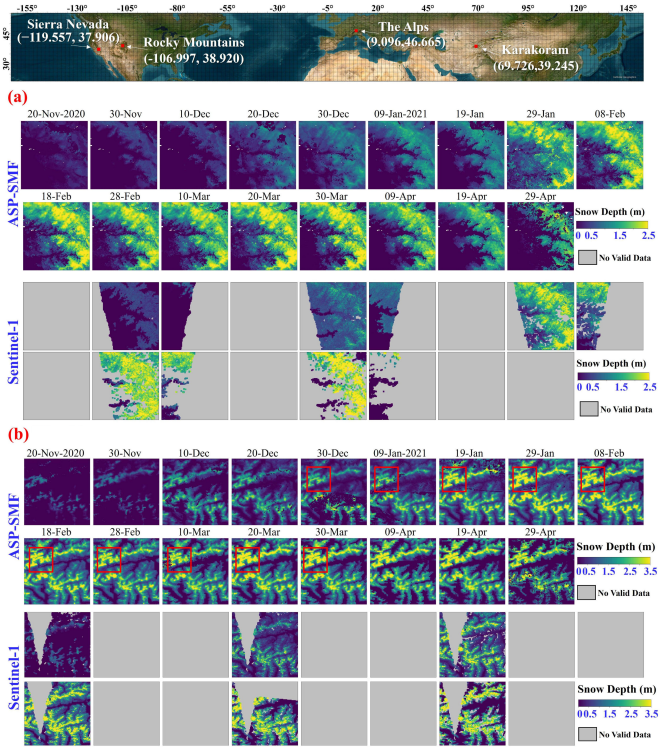


Fig. 11. Spatiotemporal evolution of snow depth from ASP-SMF and Sentinel-1 at 10-day intervals for the 2020–2021 hydrological year in (a) Sierra Nevada and (b) European Alps. Gray areas indicate no valid data in the Sentinel-1 record. Red boxes highlight transient snow events captured by ASP-SMF but missed by Sentinel-1. See Appendix D for corresponding results in the Rocky Mountains and the Karakoram.

the algorithm reconstructs a transient accumulation and melt cycle entirely absent from the sparse S1 record. Finally, this robust performance is consistent across diverse regions, from the maritime Sierra Nevada to the data-scarce Karakoram

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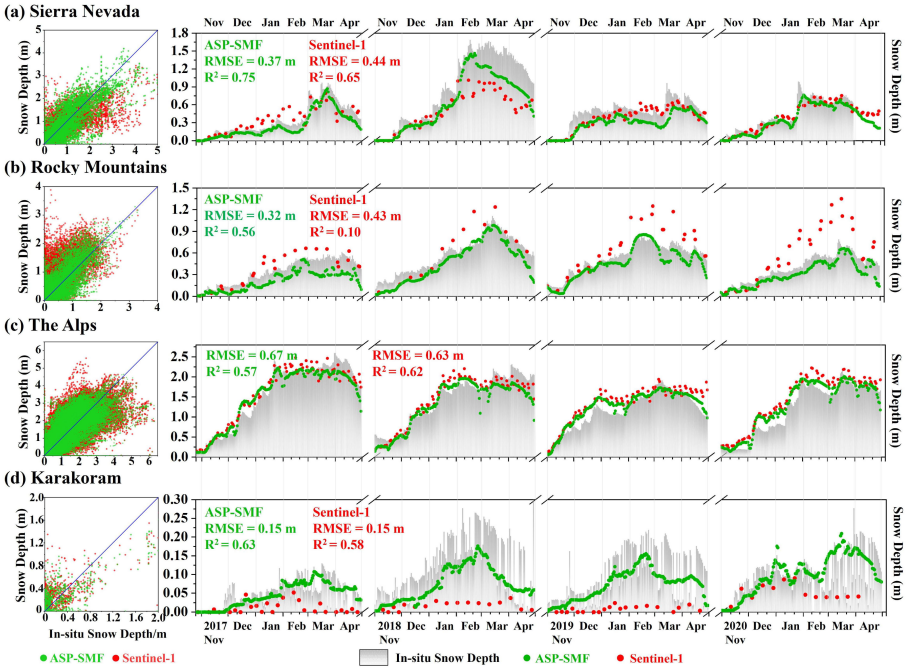


Fig. 12. Long-term (2017–2021) regional evaluation of ASP-SMF and Sentinel-1 against in situ observations. (a)–(d) For each mountain range, (left) scatter plot of all daily co-located data points and (right) time series of the regionally averaged daily snow depth.

(see Appendix D), highlighting the global applicability and generality of the ASP-SMF algorithm.

These results confirm that ASP-SMF not only fills observational gaps but also reconstructs a more complete and physically coherent temporal evolution of the mountain snowpack than is possible with SAR data alone.

2) Long-Term Regional Accuracy and Stability: To assess the long-term performance and temporal stability of the ASP-SMF product, we compared regionally averaged daily snow-depth time series against Sentinel-1 and in situ observations over the full four-year study period (see Fig. 12). The aggregated scatter plots (see Fig. 12, left panels) confirm that the daily, seamless ASP-SMF product achieves accuracy comparable to or surpassing that of the sparse Sentinel-1 observations over the long term. In the Rocky Mountains, for example, ASP-SMF (RMSE = 0.32 m and R^2 = 0.56) demonstrates a marked improvement over Sentinel-1 (RMSE = 0.43 m and R^2 = 0.10). In the Alps, the two products show comparable performance.

The regionally averaged time-series plots (see Fig. 12, right panels) further highlight the algorithm's capabilities. The continuous ASP-SMF time series (green) closely tracks the seasonal evolution depicted by the in situ data (gray background) and successfully fills the temporal gaps of Sentinel-1 (red points). The algorithm effectively mitigates the persistent overestimation observed in the Sentinel-1 data in the Rocky Mountains, a correction enabled by the RMA-SLSQP optimization. Conversely, a slight underestimation is observed in the Sierra Nevada during peak accumulation periods. In the Karakoram, all satellite-derived products struggle to consistently capture the highly dynamic and often shallow snowpack evolution characteristic of the region, highlighting a limitation of remote sensing approaches in such environments. Collectively, this long-term analysis validates ASP-SMF's ability to produce a reliable, daily, spatially seamless 500-m snow-depth product.

V. Discussion

A. Justification and Validation of the Snow Phase Classification

A key design feature of the ASP-SMF algorithm is the classification of the snow season into distinct phases. This approach acknowledges that snowpack evolution is not a monolithic process but a composite of periods dominated by different physical mechanisms. A segmented, phase-based correction model is therefore inherently more adaptable to the state-dependent error characteristics of the input data than any single, season-long fitting strategy. Here, we justify and validate the two-step classification process detailed in Section III.

The first step, initial classification, relies on deriving regionally adaptive thresholds to distinguish between stable and dynamic periods. This is crucial because a fixed threshold would fail across different mountain climates. We demonstrate this data-driven process using the Rocky Mountains as a representative example [see Fig. 13(a)]. The "knee points" detected by the Kneedle algorithm on the ECDF of 3-day snow-depth change rates mark the optimal transition from high-frequency, low-magnitude background events to low-frequency, high-impact dynamic events. For the Rockies, this corresponds to an accumulation threshold of 0.043 m/day, a point marking the 90.2% cumulative frequency on the ECDF, which separates the most intense 9.8% of accumulation events from the more frequent, smaller events. The melt threshold is

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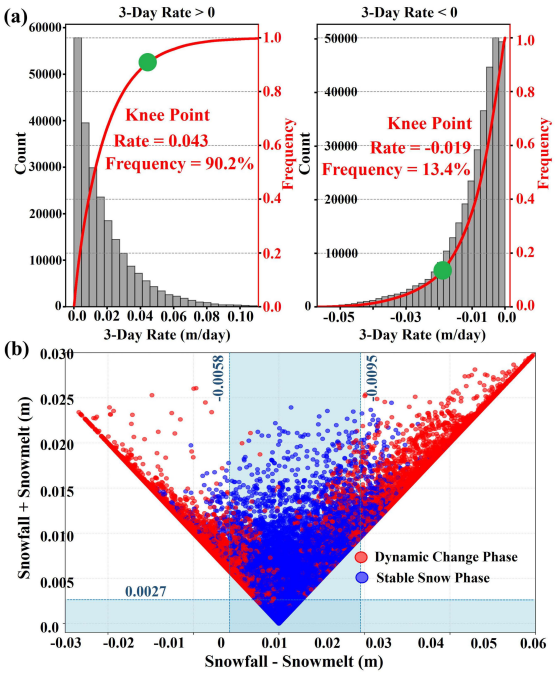


Fig. 13. Derivation and physical basis of the snow phase classification for the Rocky Mountains (2017–2021). (a) Histograms and ECDFs for (left) positive and (right) negative 3-day snow-depth change rates. Green dots mark the "knee points" identified by the Kneedle algorithm to define thresholds. (b) Feature space defined by net mass flux (*Snowfall* − *Snowmelt*) and total mass flux (*Snowfall* + *Snowmelt*). Points are colored by their final consolidated phase (stable or dynamic). The shaded area represents an example decision boundary for the consolidation algorithm.

similarly identified at −0.019 m/day, a point that delineates the most extreme 13.4% of negative change events. The physical meaning of this transition validates our use of the Kneedle algorithm to define the boundaries objectively and reproducibly between the stable and rapid phases.

The second step, phase consolidation, addresses the fragmented nature of the initial classification. A key decision is the enforcement of a six-day minimum duration for each phase. This is justified by both algorithmic and physical requirements: algorithmically, it provides sufficient data points to define a meaningful shape for fitting; physically, it filters out transient noise (e.g., diurnal melt-freeze cycles) and aligns with the typical timescale of synoptic weather systems that drive persistent snowpack changes. To further validate the physical basis of the final consolidated phases, we examined their distribution in a feature space defined by net mass flux (*Snowfall* − *Snowmelt*) and total mass flux (*Snowfall*+*Snowmelt*) [see Fig. 13(b)]. The results show a clear and physically meaningful separation: stable snow phases (blue dots) are overwhelmingly concentrated near the origin, corresponding to periods of low net change and low overall activity. In contrast, dynamic change phases (red dots) occupy the arms of the characteristic "V-shaped" distribution, indicating periods of significant mass and energy exchange. The shaded area in Fig. 13(b) exemplifies the data-driven decision boundary used by the iterative consolidation algorithm (see Appendix A) to reclassify ambiguous short segments based on these physical properties.

The effectiveness of this entire two-step process is illustrated at a representative site in the Rocky Mountains (see Fig. 14). The bottom panel for each water year shows the initial, fragmented classification based on the adaptive thresholds applied to the ERA5-Land 3-day change rate. A close inspection reveals that the periods identified as rapid accumulation or rapid melt (colored bars) align remarkably well with the actual rapid change events observed in the high-fidelity in situ and SNODAS time series (top panels). This confirms the fidelity of the ERA5-Land-derived trends; although its absolute snow-depth values may be biased, its temporal evolution is physically realistic. This is because it is driven by the HTESSEL land surface model, which simulates snowpack changes in direct physical response to daily meteorological forcings like precipitation and temperature, accurately capturing the timing of major accumulation or ablation events.

The middle panels then demonstrate the success of the consolidation algorithm, showing how the "fragmented" initial phases are systematically merged into coherent, continuous blocks of stable snow and dynamic change. Crucially, these final consolidated phases also align well with the observed snowpack evolution, clearly delineating periods of slow, stable change from the periods of rapid accumulation and melt seen in the in situ data. This robust, physically grounded segmentation provides the necessary foundation for the state-dependent bias correction, enabling the algorithm to apply the most appropriate correction strategy to each distinct stage of the snow season's evolution.

B. Addressing Input Data Uncertainty: The Critical Role of RMA-SLSQP

A fundamental challenge in this study is the fusion of two inherently imperfect datasets: a physically based model reanalysis (ERA5-Land) and a remote sensing retrieval data (Sentinel-1). The RMA-SLSQP component was specifically designed to address this challenge by acknowledging these dual uncertainties and enforcing physical constraints. To explicitly quantify the impact of this component, we conducted a comparative analysis to quantify the impact of the full ASP-SMF algorithm against a version that replaces the RMA-SLSQP optimization with a traditional OLS regression.

The results, presented in Fig. 15, show a dramatic and consistent improvement in performance when the RMA-SLSQP optimization is used. Across all four mountain ranges, the full ASP-SMF model yields substantially lower errors and higher correlations. In the Sierra Nevada, for example, its inclusion reduces the RMSE from 0.62 to 0.39 m and increases the R^2 from 0.28 to 0.72. Crucially, the OLS approach ("Without RMA-SLSQP") is prone to producing significant, nonphysical overestimations, a behavior that is effectively suppressed by our proposed model.

The reason for this performance gap lies in the flawed assumption of OLS, which treats the Sentinel-1 anchor data as infallible "ground truth." In reality, and as highlighted in the Introduction, S1 retrievals are subject to considerable uncertainty. This uncertainty stems primarily from the inherently unstable relationship between C-band backscatter and snow

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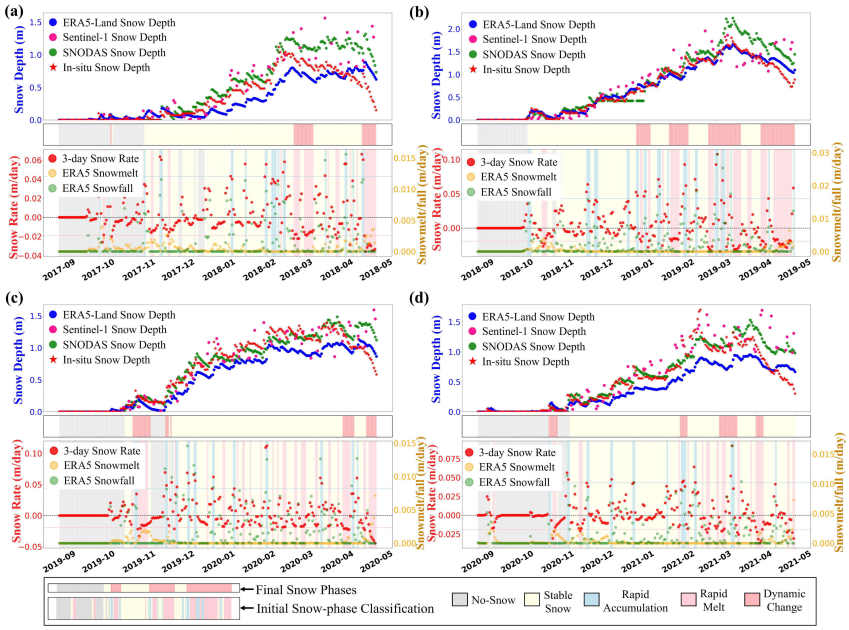


Fig. 14. Example of the two-step classification at site GHCN-USS0007K01S (Rocky Mountains) for four snow seasons: (a) September 1, 2017–April 30, 2018; (b) September 1, 2018–April 30, 2019; (c) September 1, 2019–April 30, 2020; and (d) September 1, 2020–April 30, 2021. For each panel, the top subpanel shows the snow-depth time-series comparison, the middle subpanel shows the initial four-state classification based on ERA5-Land change rates, and the bottom subpanel shows the final three-state consolidated phases used for SMF.

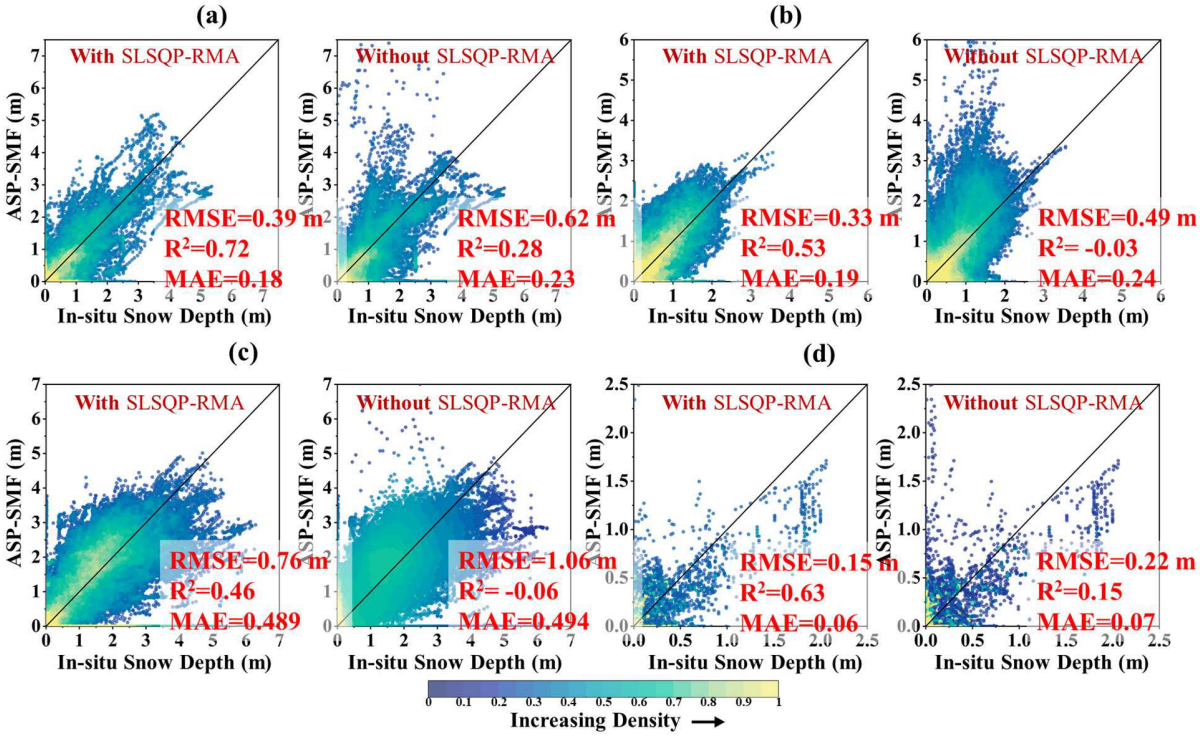


Fig. 15. Comparison study demonstrating the impact of the RMA-SLSQP component on ASP-SMF performance across the four mountain ranges (2017–2021). In each pair, (left) "With RMA-SLSQP" shows the result from the complete algorithm and (right) "Without RMA-SLSQP" shows the result when replacing the uncertainty-aware optimization with a traditional OLS regression. Note the significant reduction in RMSE when using the proposed method. (a) Sierra Nevada. (b) Rocky Mountains. (c) Alps. (d) Karakoram.

depth (particularly for shallow snow) and the empirical structure of the retrieval algorithm, which can struggle to robustly convert the SAR signal into a physical depth across varying conditions [24]. When a traditional regression encounters such an erroneously high S1 anchor point, it incorrectly scales the entire ERA5-Land time series upward to match, leading to the large biases seen in the results.

The RMA-SLSQP component mitigates this problem through a dual mechanism. First, RMA regression, by treating both datasets as uncertain, does not place undue trust in outlier

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S1 anchors. It finds a more balanced solution that respects the physically based trend from ERA5-Land, preventing the propagation of large, isolated errors. Second, the SLSQP constraints act as a physical "guardrail," preventing the amplitude parameter (α) from reaching unrealistic values during optimization. This dual mechanism is therefore essential for maintaining the algorithm's robustness and physical realism, especially in challenging mountain environments where input data quality is spatially variable.

C. Robustness to Reanalysis Input and the Choice of ERA5-Land

A critical choice for the ASP-SMF algorithm is the selection of the background reanalysis dataset that provides the temporal "shape." Recent literature has highlighted significant challenges with reanalysis snow products in mountain regions, noting that both ERA5 and ERA5-Land can exhibit substantial high biases [30]. Counterintuitively, some studies have found that the higher resolution ERA5-Land (9 km) can produce even larger and more unrealistic snow-depth values than the coarser ERA5 (31 km), particularly in the Alps [33]. This paradox motivated a sensitivity analysis to assess the impact of these two different inputs on the final ASP-SMF product.

We conducted this analysis in European Alps, comparing the results when driving the ASP-SMF algorithm with either ERA5-Land or ERA5. The analysis reveals two key findings.

First, while the raw inputs differ significantly, the final 500-m ASP-SMF outputs are remarkably similar in spatial detail. As shown in Fig. 16, the raw ERA5-Land data can exhibit anomalous "hot spots" of extreme snow depth that are inconsistent with the surrounding topography (e.g., the patch highlighted in Fig. 16(a), region of interest (ROI) 1). Crucially, these artifacts are entirely removed by the ASP-SMF algorithm, which produces a physically plausible and natural snow distribution that aligns with the fine-scale terrain, regardless of the input (see Fig. 16(c), red circle).

Second, the quantitative evaluation confirms this robustness (see Fig. 17). Despite the poor performance and divergent biases of the raw inputs (ERA5-Land: RMSE = 2.43 m, large positive bias; ERA5: RMSE = 0.92 m, large negative bias), the final ASP-SMF products are of a similarly high quality, with RMSEs of 0.64 m (using ERA5-Land) and 0.66 m (using ERA5).

This pronounced robustness stems from the fundamental design of the ASP-SMF correction model. The algorithm is intentionally sensitive to the temporal pattern ("shape") of the reanalysis data within each classified phase but is designed to be insensitive to its absolute magnitude and offset. These biases are explicitly corrected by the scaling (α) and offset (γ) parameters during the phasewise fitting process [see (1)]. As long as both ERA5 and ERA5-Land provide a physically plausible temporal sequence of accumulation and melt events, the algorithm can successfully rescale and shift this pattern to match the Sentinel-1 anchors. This demonstrates that the fidelity of the temporal evolution, rather than absolute values or the native spatial resolution of the reanalysis, is the critical requirement for the input data in the ASP-SMF algorithm. Accordingly, the scale mismatch between the reanalysis background and the 500-m output grid is not eliminated by interpolation, but its influence is reduced because the reanalysis mainly provides the temporal shape, while Sentinel-1 and IMS provide the high-resolution observational constraints.

Given the similar performance, our choice of ERA5-Land as the primary input was based on its slightly superior final accuracy in this test case and its higher native spatial resolution. We posit that the 9-km grid may offer a better representation of fine-scale orographic effects on the timing of precipitation and melt, even if the magnitude is more biased than the 31-km ERA5 product.

D. Implications, Limitations, and Future Directions

A major implication of this work is the potential for ASP-SMF to serve as a globally scalable, in situ-independent alternative to data-intensive operational products like SNODAS. While SNODAS provides high-quality estimates, it is geographically limited to the contiguous U.S., computationally demanding, and reliant on a dense network of ground observations for assimilation. In contrast, ASP-SMF is computationally efficient and relies exclusively on globally open-access satellite and reanalysis data. Our results demonstrate that it achieves performance levels often comparable—and in some cases superior—to SNODAS [e.g., a higher KGE in the Rocky Mountains, Fig. 6(b)]. A direct comparison between the two products (see Fig. 18) further confirms their strong agreement (e.g., $R^2 = 0.77$ in the Sierra Nevada), suggesting that ASP-SMF can provide a valuable, high-resolution (500 m) daily snow-depth dataset for the vast, ungauged mountain ranges of the world.

Despite its strong performance, the ASP-SMF algorithm has several limitations that open avenues for future research.

First, while the algorithm is designed to be robust against biases, its final accuracy is inevitably conditioned by the quality of its inputs, particularly the Sentinel-1 anchors (see Fig. 10). This dependency explains the lower accuracy observed in the Karakoram, where extreme topography causes severe SAR geometric distortions (layover and shadow) and complex scattering physics in cold, wind-scoured snowpacks challenge the empirical retrieval algorithm. Similarly, this dependency likely accounts for the underperformance in European Alps in 2019 (see Section IV-B), where an unusually persistent low bias in the Sentinel-1 input could not be fully overcome even by the robust RMA-SLSQP optimization. Forest canopy is another important source of uncertainty for C-band SAR snow-depth retrievals because vegetation can attenuate and occlude the radar signal. We therefore added a canopy-stratified dry-snow validation in Appendix E. The results show that ASP-SMF improves over Sentinel-1 across all canopy-fraction classes, but the reduced correlation and KGE under the densest canopy conditions indicate that closed-canopy forests remain an important operational limitation.

Second, although the algorithm handles moderate data sparsity well, performance may degrade if anchor points become extremely sparse over prolonged periods, leaving the ERA5-Land trend under-constrained. However, this limitation is expected to diminish with the launches of the Sentinel-1C

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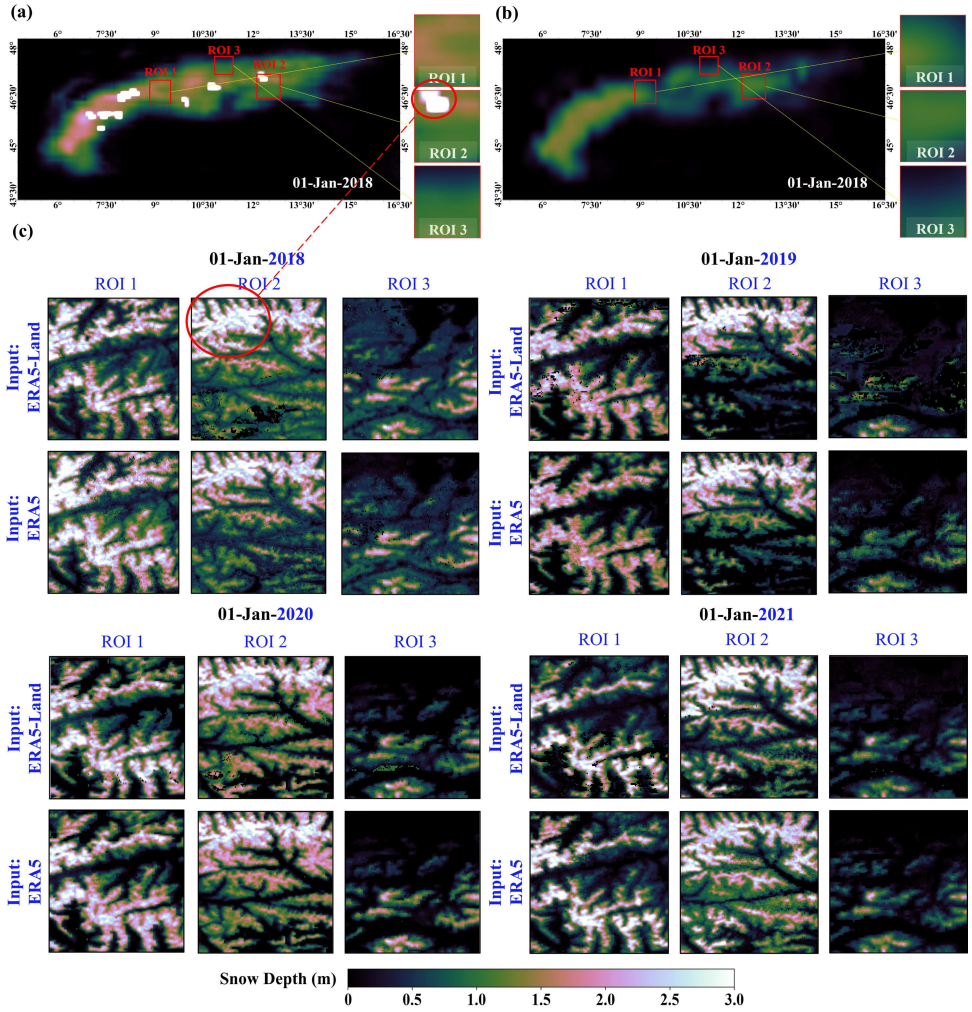


Fig. 16. Visual comparison of ASP-SMF results using ERA5-Land versus ERA5 as input in the Alps. (a) and (b) Raw snow depth from ERA5-Land (9 km) and ERA5 (31 km) for January 1, 2018, with three ROIs. (c) Comparison of the final 500-m ASP-SMF product generated using either ERA5-Land or ERA5 as input for the three ROIs across four different years. The red circle highlights the successful removal of an anomalous "hot spot" from the raw ERA5-Land data in the final product.

and Sentinel-1D satellites, which will significantly improve temporal revisit frequency and data availability.

Third, as a pixelwise method, ASP-SMF processes each grid cell independently, which can occasionally result in salt-and-pepper "speckle" artifacts in the final maps [see Fig. 6(a)]. Future versions could explicitly incorporate spatial context to improve map coherence, for example, through Markov random field-type spatial regularization or graph-based smoothing [65], object- or superpixel-based processing [66], and topographically constrained "similar-pixel" strategies that borrow information from neighboring pixels with comparable terrain and snow-season characteristics [51]. These approaches could be used either as postprocessing constraints or within the fitting stage to provide supplementary anchors during data-poor periods, while preserving terrain-driven snow-depth gradients.

Fourth, future iterations of ASP-SMF could further reduce the influence of flawed Sentinel-1 anchors by incorporating topography-aware confidence weighting. This is particularly relevant for high-relief regions such as the Karakoram, where terrain-induced layover, radar shadow, and unfavorable SAR illumination geometry can degrade Sentinel-1 snow-depth retrievals. DEM-derived terrain-geometry metrics, including slope, aspect relative to the radar look direction, local incidence angle, and layover/shadow risk masks, could be used to estimate the reliability of each Sentinel-1 anchor. Anchors in severe layover or radar-shadow areas could be screened out, whereas those in less favorable but still usable geometries could receive lower weights during RMA-SLSQP fitting. Such an extension would preserve reliable high-resolution SAR constraints while reducing the impact of erroneous observations on the fit correction parameters.

Finally, while the in situ-independent nature of ASP-SMF is a key strength, its output could serve as a high-quality baseline background field for assimilating local station data where available, potentially producing a highly accurate regional product. Furthermore, the principles of the ASP-SMF algorithm are sensor-agnostic and could be readily adapted to upcoming missions like NISAR (NASA-ISRO SAR). By integrating the high-resolution, InSAR-based snow-depth retrievals from NISAR with the temporal continuity of

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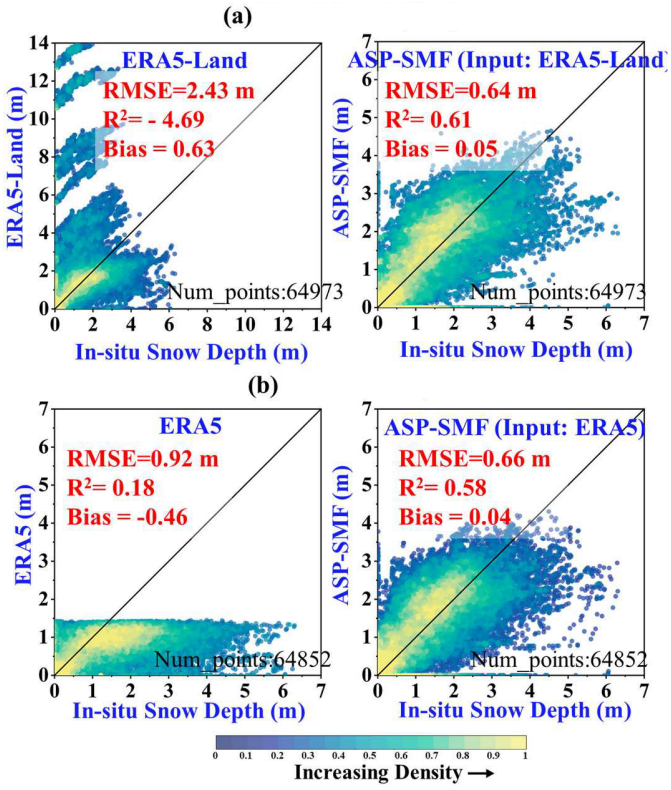


Fig. 17. Quantitative comparison of ASP-SMF results using ERA5-Land versus ERA5 in the Alps. (a) Scatter plots showing the performance of (left) raw ERA5-Land (9 km) and (right) resulting ASP-SMF product against in situ data. (b) Same analysis for (left) raw ERA5 (31 km) and (right) resulting ASP-SMF product.

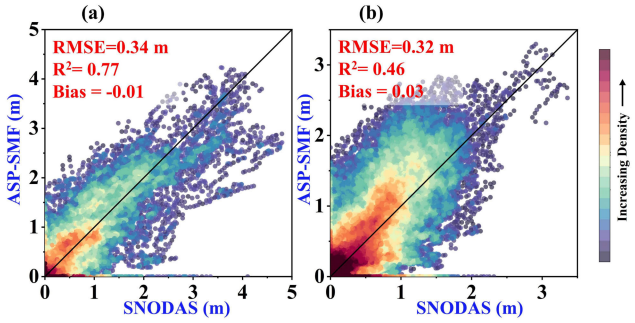


Fig. 18. Direct comparison of daily ASP-SMF and SNODAS snow depth for all co-located pixels from September 2017 to April 2021 in (a) Sierra Nevada and (b) Rocky Mountains. The high correlation demonstrates the potential of ASP-SMF as a scalable complement to regionally limited snow data assimilation products such as SNODAS.

reanalysis, the ASP-SMF approach can be extended to create even more powerful tools for global snow monitoring, enabling high-resolution, long-term assessments of snow water resources in a changing climate.

VI. Conclusion

Accurately quantifying daily snow depth across remote, complex mountain terrain remains a major challenge in Earth observation. While high-resolution SAR offers significant potential, its operational utility is fundamentally constrained by infrequent revisits and a loss of sensitivity in wet snow, preventing the capture of dynamic snowpack processes. This study proposed and validated the ASP-SMF algorithm, a novel physically constrained fusion algorithm designed to generate daily, seamless, 500-m snow-depth maps. By synergizing globally available public datasets (Sentinel-1, ERA5-Land, and IMS) independently of in situ calibration, ASP-SMF achieves a robust and scalable solution for Northern Hemisphere mountains. This is achieved through a two-stage process that first adaptively segments the snow season into distinct physical phases and then applies an uncertainty-aware optimization (RMA-SLSQP) to correct for temporal, amplitude, and offset biases in the reanalysis data.

Comprehensive validation against airborne LiDAR, photogrammetry, and in situ measurements demonstrated three key findings. First, the daily seamless ASP-SMF algorithm consistently achieves an accuracy that is comparable or superior to its sparse Sentinel-1 anchor data. Relative to the original Sentinel-1 retrievals, ASP-SMF reduced point-scale RMSE by 26% under dry-snow conditions, from 0.27 to 0.20 m, and increased R^2 from 0.49 to 0.72. Second, and most critically, the algorithm effectively addresses the primary limitation of C-band SAR by maintaining high accuracy during wet snow conditions, boosting R^2 by 145% compared with original Sentinel-1 retrievals. Third, despite being a globally applicable and in situ-independent method, ASP-SMF achieved performance comparable to the data-assimilating SNODAS product, with a higher KGE in the Rocky Mountains. Viewed against the regional SAR-, ML-, and fusion-based snow-depth retrieval benchmarks summarized in Section IV-C, these results indicate that ASP-SMF provides competitive accuracy while offering a distinct practical advantage: it converts sparse and wet-snow-sensitive Sentinel-1 retrievals into a daily, spatially seamless, 500-m snow-depth product across diverse mountain regions without in situ calibration. Therefore, the main advancement of ASP-SMF lies not only in improving retrieval accuracy, but also in enabling temporally continuous high-resolution snow-depth monitoring for data-scarce mountain environments.

As an open-source and computationally efficient algorithm based entirely on public data, ASP-SMF offers a scalable pathway to generate daily, 500-m snow-depth records for the world's vast, ungauged mountain ranges. It thus provides the cryospheric and hydrological communities with a critical new tool, creating opportunities to improve water resource management, enhance hydrological modeling, and advance hazard assessment in vital mountain water towers under climate change.

Appendix A
Iterative Temporal Consolidation Algorithm

A. Objective and Overall Procedure

The iterative temporal consolidation algorithm resolves the temporal fragmentation of the initial daily classification (see Section III-A). It systematically merges or reclassifies all contiguous phase segments shorter than a minimum duration (τ_d = 6 days) to produce a continuous and physically coherent time series. The algorithm operates through an iterative loop, processing segments of increasing length from $N = 1$ to $N = 5$

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days. In each iteration N, all segments of that specific length are identified and reclassified based on a hierarchical set of decision rules, ensuring that the shortest, most ambiguous segments are resolved first.

B. Hierarchical Decision Rules for Segment Reclassification

For each target segment of length N, a two-rule hierarchy is applied. Rule 1 is always evaluated first; Rule 2 is invoked only if the conditions for Rule 1 are not met.

1) Rule 1 (Contextual Majority Assimilation): This rule leverages a segment's temporal context to smooth out likely classification noise. A dynamic "contextual window" (W_C) of length $3N$ is centered on the target N-day segment. If any single phase within this window (excluding the target segment) has a total duration of $N+1$ days or more, it is considered the dominant phase, and the target segment is assimilated into it. This rule is based on the principle that an isolated, transient phase is more likely to be an outlier within a persistent, stable neighborhood.

2) Rule 2 (Conditional Physical Re-Evaluation): If contextual assimilation is not possible, this rule uses physical variables from ERA5-Land to determine the most plausible classification. The evaluation is based on the mean values of snow depth ($\overline{SD}$), net mass flux ($\overline{F_{net}} = \overline{Snowfall - Snowmelt}$), and total mass flux ($\overline{F_{sum}} = \overline{Snowfall + Snowmelt}$) within a local window around the target segment. The logic is conditional upon the segment's current classification.

For a segment initially classified as no-snow phase, the re-evaluation determines if it represents a stable state or a brief transition. If the mass exchange is low, indicated by the net flux falling within a near-zero range ($T_{net,low} < \overline{F_{net}} \leq T_{net,high}$) or the total flux being below a stability threshold ($\overline{F_{sum}} \leq \tau_{sum}$), the segment is reclassified to stable snow phase. Otherwise, significant underlying flux suggests an imminent change, and the segment is reclassified to the dynamic change phase.

For a segment classified as stable snow phase, the rule checks for misclassifications due to very shallow snow. If the mean snow depth ($\overline{SD}$) falls below the commonly utilized binary snow detection threshold (τ_{sd} = 0.02 m) [67], [68], the segment is physically more representative of snow-free conditions and is reclassified to no-snow phase. Otherwise, its classification is retained.

Finally, for a segment classified as the dynamic change phase, the rule acts as a physical plausibility filter. A significant dynamic event is implausible without a substantial snowpack; therefore, if $\overline{SD} \leq \tau_{sd}$, the segment is corrected to no-snow phase. In addition, if the segment passes the depth check but was not contextually assimilated (Rule 1), it is considered an isolated artifact rather than a persistent physical process. Consequently, it is reclassified to the more plausible stable snow phase to prevent spurious rapid fluctuations.

Upon completion of the N = 1–5 loop, all phase segments have a duration of at least six days.

C. Algorithm Parameters

The key parameters and thresholds used in the algorithm are summarized in Table III. While some parameters like τ_d and τ_{sd} are fixed, the crucial mass flux thresholds used in Rule 2 are regionally adaptive to account for varying climatic characteristics.

TABLE III
PARAMETERS FOR THE ITERATIVE TEMPORAL CONSOLIDATION ALGORITHM

Params	Description	Value / Definition
τ_d	Minimum final duration of a phase segment	6 days
N	Length of the target segment being processed	Iterates from 1 to 5
W_C	Size of the Contextual Window for Rule 1	$3N$ days
τ_{sd}	Minimum snow depth threshold	0.02 m
$\tau_{net,low}$	Lower bound for net mass flux threshold	Regionally adaptive (see text)
$\tau_{net,high}$	Upper bound for net mass flux threshold	Regionally adaptive (see text)
τ_{sum}	Upper bound for total mass flux threshold	Regionally adaptive (see text)

These adaptive thresholds are derived from the 2-D distribution of net mass flux (F_{net}) versus total mass flux (F_{sum}) for each mountain range. As illustrated in Fig. 13, the data points form a characteristic "V-shaped" distribution. The stable snow phase points cluster near the origin (low F_{net} and low F_{sum}), while dynamic change phase points occupy the upper arms of the "V." The thresholds ($\tau_{net,low}, \tau_{net,high}, \tau_{sum}$) are defined to delineate the boundary of this stable, low-activity region near the origin. This data-driven approach ensures that the physical re-evaluation rules are tailored to the specific snowfall and melt intensities of each region.

APPENDIX B
ASSESSMENT OF SENTINEL-1 OBSERVATION FREQUENCY

To evaluate the data availability for snow monitoring, we calculated the average monthly Sentinel-1 observation frequency over the Sierra Nevada (2017–2021). We compared "nominal observations" (all satellite overpasses) with "effective observations," which exclude pixels flagged as wet snow provided by Lievens et al. [6], because liquid water absorbs C-band signals and precludes reliable snow-depth retrieval. Results indicate that while the nominal revisit frequency is 6–12 times per month [see Fig. 19(a)], wet snow masking notably reduces the number of valid observations [see Fig. 19(b)]. Specifically, in April, effective observations in high-elevation areas frequently drop to fewer than two per month [see Fig. 19(c)], limiting the temporal resolution required to capture daily snow dynamics during the melt season.

APPENDIX C
INTEGRATION AND VALIDATION OF IMS SNOW-FREE ANCHORING

The integration of IMS snow cover data serves a critical role in supplementing the sparse Sentinel-1(S1) retrievals.

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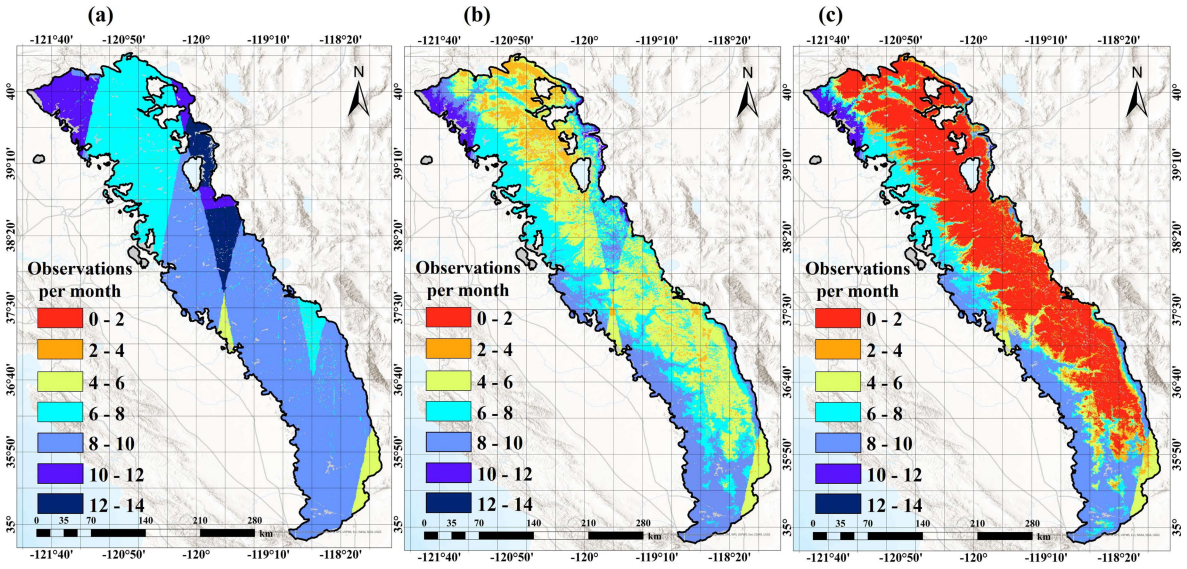


Fig. 19. Average monthly Sentinel-1 observation frequency in the Sierra Nevada (2017–2021). (a) Nominal monthly overpasses derived from the Sentinel-1A/B orbit. (b) Effective observations per month after removing wet snow pixels (October–April average). (c) Effective observations per month in April, highlighting data loss during peak melt.

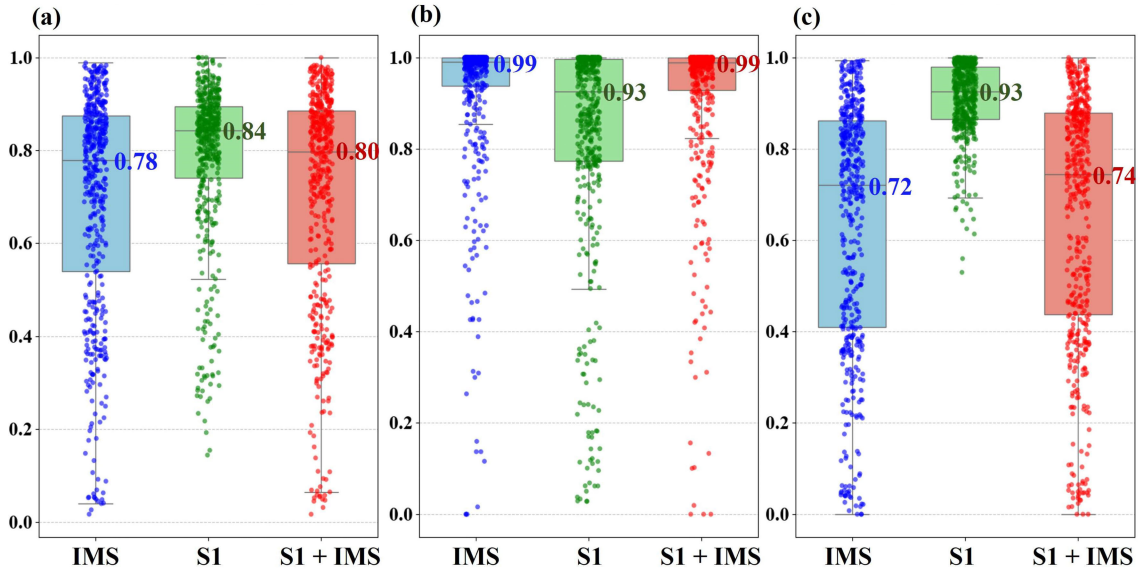


Fig. 20. Site-level validation of the binary snow detection performance of IMS, Sentinel-1 (dry snow conditions only), and the combined S1 + IMS anchor dataset. Box plots show the distribution of (a) accuracy, (b) precision, and (c) recall scores calculated against all in situ station observations (2017–2021).

Specifically, on days where valid Sentinel-1 data are unavailable (due to revisit cycles or wet snow masking) but IMS identifies the pixel as snow-free, we explicitly assign a snow depth of zero ($SD = 0$) to the anchor dataset.

This strategy offers two key advantages. First, it significantly expands the volume of available anchor points, improving the robustness of the SMF. Second, and most critically, it provides essential constraints during the wet snow phase. During late spring melt, C-band SAR signals are often attenuated by liquid water, leading to data gaps; however, IMS continues to reliably detect the emergence of bare ground. By utilizing IMS to pin the snow depth to zero immediately upon melt-out, the algorithm prevents the model from erroneously extending the snow season.

To validate this integration, we assessed the binary snow detection performance of IMS, Sentinel-1, and their combination against in situ station data (see Fig. 20). The analysis reveals complementary strengths: Sentinel-1 exhibits high recall (0.93), ensuring existing snow is detected, while IMS achieves near-perfect precision (0.99). This implies that when IMS classifies a pixel as "snow-free," it is highly reliable.

Crucially, while the combined S1 + IMS product may show slightly lower metric scores compared with Sentinel-1 alone in some instances (e.g., lower recall due to IMS's coarser resolution), this comparison must be contextualized by

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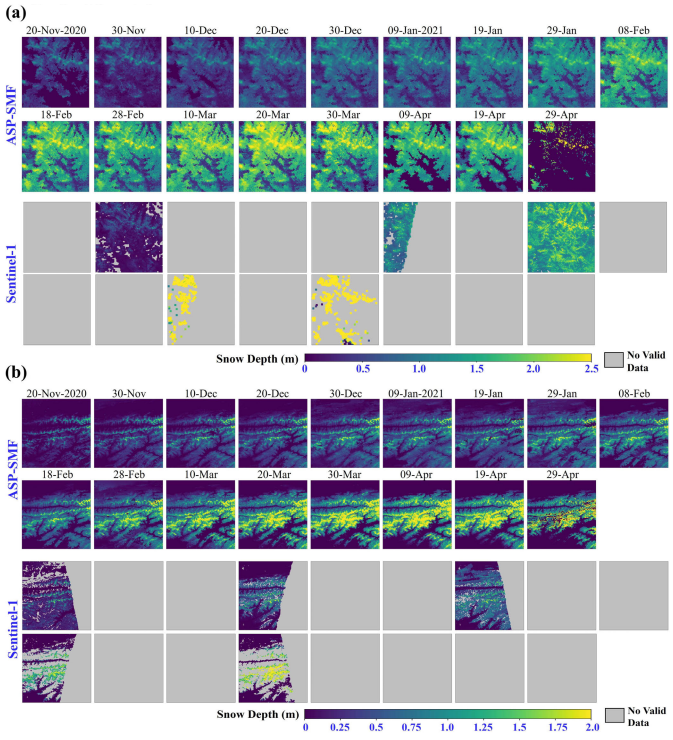


Fig. 21. Spatiotemporal evolution of snow depth from ASP-SMF and Sentinel-1 at 10-day intervals for the 2020–2021 water year in (a) Rocky Mountains and (b) Karakoram. Gray areas indicate no valid data in the Sentinel-1 record.

data availability. The Sentinel-1 metrics are calculated only on the sparse subset of days with valid satellite overpasses. In contrast, the S1 + IMS product provides a far more comprehensive temporal coverage, filling critical gaps during wet snow periods. Therefore, the combined product represents a robust compromise: it leverages the high spatial resolution of S1 where available, while using the high-precision zero-snow constraint of IMS to anchor the timeline where S1 fails, ultimately preventing the shape model from predicting false snow cover in the absence of SAR data.

Appendix D
Spatiotemporal Evolution in Continental and Data-Scarce Regions

This appendix presents the spatiotemporal evolution of snow depth derived from the ASP-SMF algorithm for the Rocky Mountains and the Karakoram, complementing the analysis in Section IV-E Fig. 21 illustrates the daily, seamless 500-m snow-depth maps at 10-day intervals for the 2020–2021 water year, alongside available Sentinel-1 retrievals.

In the Rocky Mountains [see Fig. 21(a)], ASP-SMF effectively captures the widespread, persistent continental snowpack, filling extensive temporal gaps in the Sentinel-1 record while preserving the spatial detail of snow distribution in complex terrain. In the Karakoram [see Fig. 21(b)], the algorithm demonstrates its robustness in a data-scarce environment, reconstructing coherent snow cover dynamics over rugged topography where ground observations are virtually nonexistent, and satellite revisits are infrequent. These results further validate the global generalizability of the ASP-SMF framework to regions with distinct snow climates and limited observational constraints.

Appendix E
Canopy-Stratified Dry-Snow Validation

Forest canopy can attenuate and occlude C-band SAR signals, thereby affecting Sentinel-1 snow-depth retrievals and the final ASP-SMF product. To assess this effect, we performed a canopy-stratified validation using the Copernicus Global Land Service 100-m tree-cover fraction product. The tree-cover fraction data were aggregated to the 500-m ASP-SMF grid, and the validation samples were divided into five canopy-fraction classes: 0%–20%, 20%–40%, 40%–60%, 60%–80%, and 80%–100%. ERA5-Land, Sentinel-1, and ASP-SMF were evaluated against station observations within each canopy-fraction class (see Table IV and Fig. 22).

We then evaluated ERA5-Land, Sentinel-1, and ASP-SMF against station observations under dry-snow conditions for each canopy-fraction class (see Table IV). ASP-SMF shows consistently lower RMSE than Sentinel-1 across all canopy classes. Its RMSE values are 0.28, 0.22, 0.17, 0.15, and 0.15 m from the lowest to highest canopy-fraction classes, respectively, with small biases ranging from −0.02 to 0.02 m. These results indicate that ASP-SMF can partly mitigate canopy-related uncertainty by combining Sentinel-1 anchors with the temporally continuous ERA5-Land background.

The lower RMSE in high-canopy classes should be interpreted with caution. It does not imply that dense forest improves the C-band SAR snow-depth retrieval. Rather, high-canopy validation samples tend to have a narrower snow-depth range and fewer very deep snow cases, which can reduce absolute error metrics such as RMSE and MAE. In contrast, R^2, Spearman's ρ, and KGE decrease in the 80%–100% canopy class, indicating that closed-canopy forests can still weaken the temporal and rank consistency of the retrieval. Therefore, dense forest remains an important operational limitation of the current product.

Appendix F
Uncertainty Quantification and Site-Level Significance Testing

To quantify the uncertainty of the station-based validation metrics and evaluate whether the improvements of ASP-SMF were statistically significant across sites, we conducted an additional site-level statistical analysis. The analysis used the same common valid station-date samples as those used in Fig. 8.

CIs were estimated using bootstrap resampling, a standard nonparametric approach for quantifying the sampling uncertainty of statistical estimates [69], [70], [71]. Here, CI refers to CI; a 95% CI represents the interval bounded by the 2.5th and 97.5th percentiles of the bootstrap distribution. Because daily snow-depth observations from the same station are temporally correlated, individual daily observations were not resampled independently. Instead, we used a station-level bootstrap strategy consistent with cluster-bootstrap practice

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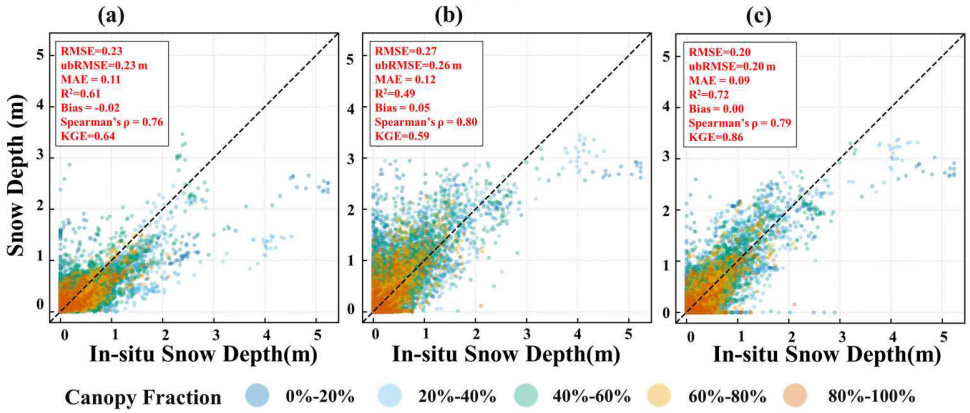


Fig. 22. Dry-snow station-based scatterplots colored by canopy-fraction class. (a) ERA5-Land. (b) Sentinel-1. (c) ASP-SMF. The colors indicate different canopy-fraction classes.

TABLE IV
CANOPY-STRATIFIED DRY-SNOW VALIDATION METRICS FOR ERA5-LAND, SENTINEL-1, AND ASP-SMF

Product	Canopy Fraction	Number of points	RMSE	ubRMSE	MAE	R^2	Bias	Spearman's ρ	KGE
ERA5-Land	0-20%	4047	0.33	0.33	0.16	0.51	0.03	0.67	0.53
	20-40%	7482	0.27	0.27	0.11	0.63	-0.05	0.78	0.54
	40-60%	9369	0.21	0.21	0.11	0.63	-0.01	0.78	0.73
	60-80%	5319	0.14	0.14	0.07	0.70	-0.02	0.78	0.68
	80-100%	1861	0.13	0.13	0.07	0.60	0.00	0.75	0.67
Sentinel-1	0-20%	4047	0.40	0.38	0.17	0.31	0.11	0.76	0.29
	20-40%	7482	0.26	0.26	0.12	0.65	0.06	0.83	0.66
	40-60%	9369	0.25	0.24	0.11	0.46	0.05	0.83	0.58
	60-80%	5319	0.20	0.20	0.09	0.44	0.02	0.79	0.64
	80-100%	1861	0.18	0.18	0.08	0.24	0.01	0.70	0.64
ASP-SMF	0-20%	4047	0.28	0.28	0.11	0.66	0.02	0.75	0.75
	20-40%	7482	0.22	0.22	0.10	0.76	0.02	0.84	0.86
	40-60%	9369	0.17	0.17	0.08	0.74	0.00	0.81	0.87
	60-80%	5319	0.15	0.15	0.07	0.65	-0.01	0.76	0.81
	80-100%	1861	0.15	0.14	0.06	0.53	-0.02	0.63	0.63

for grouped data [72]. In each bootstrap iteration, station/grid cells were resampled with replacement, and all valid daily observations within the selected sites were retained. This approach preserves the within-site temporal structure while treating the station/grid cell as the independent sampling unit. The procedure was repeated 5000 times.

Statistical significance was assessed using paired site-level Wilcoxon signed-rank tests [64]. This nonparametric paired test was selected because ASP-SMF and Sentinel-1 were evaluated at the same sites, and the site-level metric differences were not assumed to be normally distributed. The tests were applied to site-level differences in RMSE and KGE between ASP-SMF and Sentinel-1.

The results show that ASP-SMF significantly improves the station-based validation metrics under both dry and wet snow conditions (see Table V and Fig. 23). Under dry snow conditions, ASP-SMF reduced RMSE from 0.267 to 0.200 m, and the paired site-level test indicated a statistically significant improvement ($p < 0.001$). Under wet snow conditions, ASP-SMF reduced RMSE from 0.611 to 0.469 m, with the improvement also significant across sites ($p < 0.001$). KGE improvements were likewise significant under both snow conditions.

Code and Data Availability

The source code for the adaptive snow-phenology shape model fitting (ASP-SMF) algorithm, along with sample datasets and a demonstration of the processing workflow, is openly available on GitHub at https://github.com/ZhangCheng-Geo/ASP-SMF. All primary and validation datasets used in this study are publicly archived and can be accessed from their respective repositories. The daily aggregated ERA5-Land and ERA5 reanalysis datasets were obtained from the Copernicus Climate Data Store (https://cds.climate.copernicus.eu). The Sentinel-1 snow-depth product is available from the C-SNOW project data portal (https://ees.kuleuven.be/project/c-snow). The IMS data are available from the NOAA at National Snow and Ice Data Center (NSIDC) DAAC (https://nsidc.org/data/g02156). The Airborne Snow Observa-

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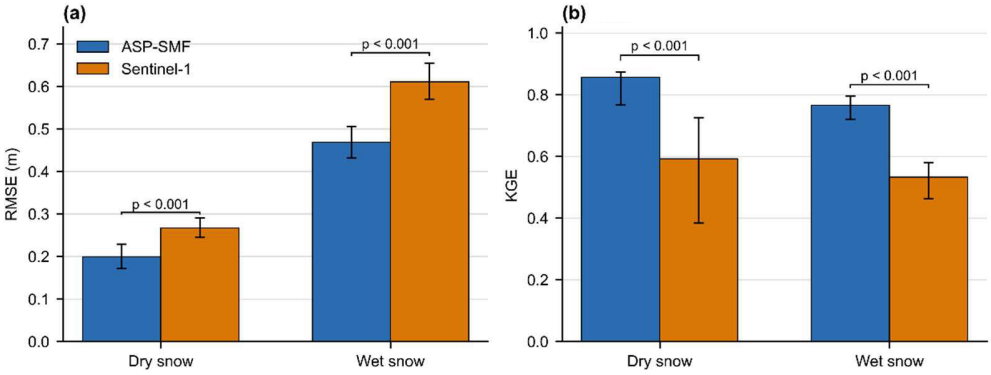


Fig. 23. Uncertainty and statistical significance of ASP-SMF improvements over Sentinel-1. Error bars indicate station-level bootstrap 95% CIs. The p-values were obtained from paired site-level Wilcoxon signed-rank tests. (a) RMSE. (b) KGE.

TABLE V

UNCERTAINTY QUANTIFICATION AND PAIRED SITE-LEVEL SIGNIFICANCE TESTS FOR ASP-SMF AND SENTINEL-1 UNDER DRY AND WET SNOW CONDITIONS. VALUES IN BRACKETS INDICATE STATION-LEVEL BOOTSTRAP 95% CIS

Snow condition	Metric	ASP-SMF (95% CI)	Sentinel-1 (95% CI)	Δ ASP-SMF − Sentinel-1 (95% CI)	Paired site-level test
Dry	RMSE	0.200 [0.172, 0.229]	0.267 [0.245, 0.291]	-0.068 [-0.086, -0.050]	$p < 0.001$
Dry	KGE	0.857 [0.768, 0.874]	0.592 [0.384, 0.726]	0.265 [0.119, 0.398]	$p < 0.001$
Wet	RMSE	0.469 [0.432, 0.506]	0.611 [0.570, 0.655]	-0.143 [-0.182, -0.104]	$p < 0.001$
Wet	KGE	0.766 [0.720, 0.796]	0.533 [0.463, 0.580]	0.233 [0.169, 0.301]	$p < 0.001$

tory (ASO) data were downloaded from the NSIDC DAAC (https://nsidc.org/data/aso). The SNOw Data Assimilation System (SNODAS) data are available from the NSIDC DAAC (https://nsidc.org/data/g02158). The airborne photogrammetry snow-depth data for Davos region are available from the EnviDat repository (https://www.envidat.ch). The GHCN data were obtained from the NOAA National Centers for Environmental Information (NCEI) (https://www.ncei.noaa.gov). The Intercantonal Measurement and Information System (IMIS) station data for the Swiss Alps were accessed from the WSL Institute for Snow and Avalanche Research SLF data portal (https://www.slf.ch/en/services/measurement-data-imis).

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